

KMOU Division of Marine System Engineering

Introduction to Internal Combustion Engine



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Lesson content for each week

Ch0_Introduction to the Class

Ch6_Piston Fittings and Crankshaft

Ch1_Overview of the Internal Combustion Engine

Ch7_Lube Oil System

Ch2_Thermodynamics and Compressed Air System

Ch8_Cooling Water System

Ch3_Marine Fuels and Internal Fuel Oil System

Ch9_Introduction to the Main Engine

Ch4_Intake and Exhaust System

Ch10_Alternative Fuels and Alternative Fuel Systems

Ch5_Turbocharger

Ch11_Environmental Regulations and Exhaust Gas Aftertreatment Systems

Midterm

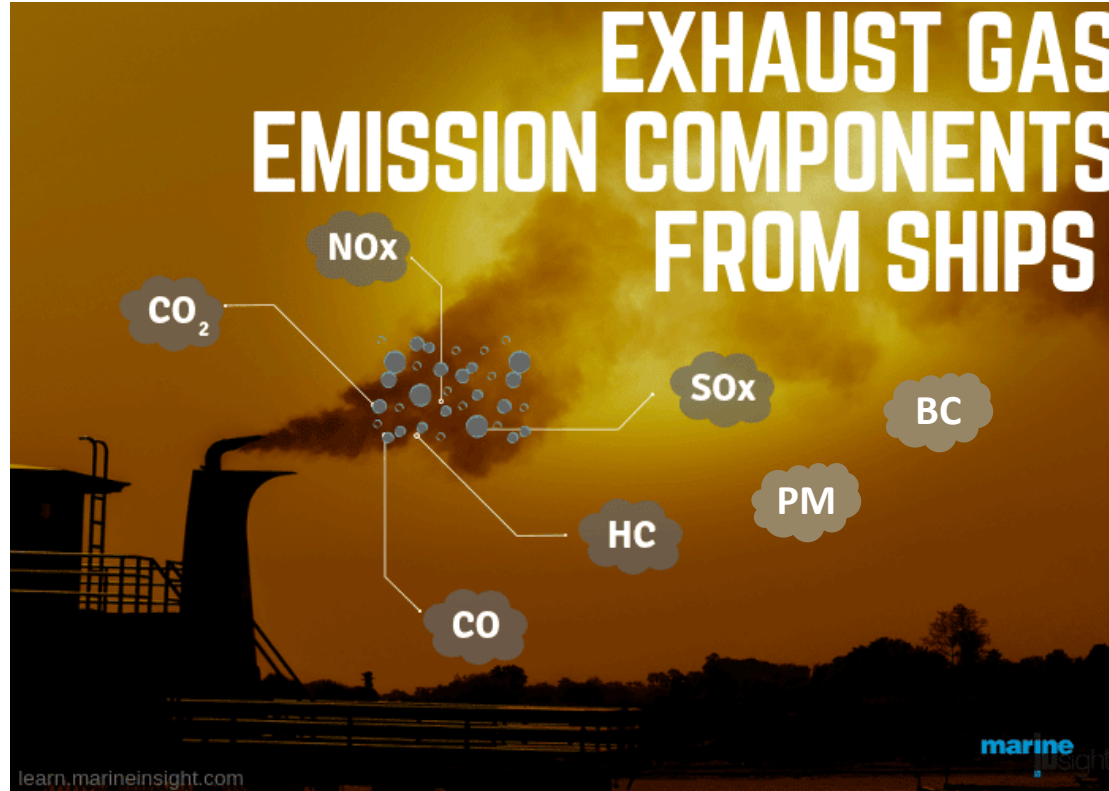
Ch12_Tools, Apprentice Engineer Duties, and Career Direction

Final Exam

Exhaust gas from ships



Exhaust gas from ships





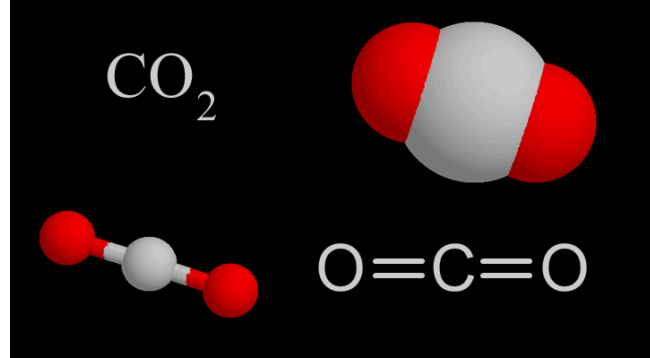
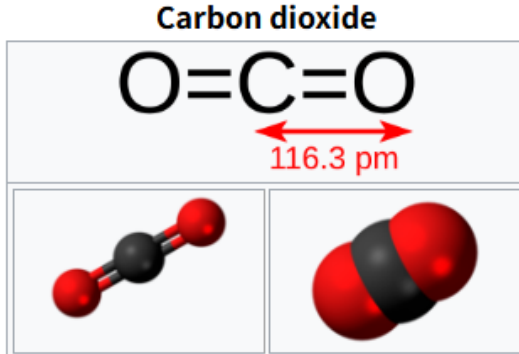
1. CO₂ (Carbon dioxide):

CO₂ is a chemical compound with the chemical formula CO₂.

It is made up of molecules that each have one carbon atom covalently double bonded to two oxygen atoms.

CO₂ is the primary cause of climate change.

In the air, CO₂ is transparent to visible light but absorbs infrared radiation, acting as a greenhouse gas.



CLIMATE CHANGE EFFECTS



HOTTER
TEMPERATURES



MORE SEVERE
STORMS



INCREASED
DROUGHT



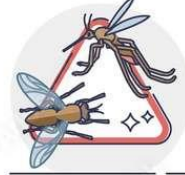
A WARMING, RISING
OCEAN



LOSS
OF SPECIES



NOT ENOUGH
FOOD



MORE
HEALTH RISKS



POVERTY AND
DISPLACEMENT

Causes and Effects of Climate Change



CO

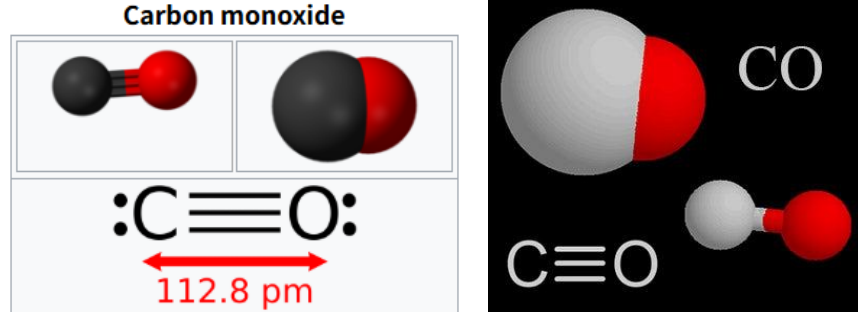
2. CO (Carbon monoxide):

CO is a poisonous, flammable gas that is colorless, odorless, tasteless, and slightly less dense than air.

CO consists of one carbon atom and one oxygen atom connected by a triple bond.

It is a key ingredient in many processes in industrial chemistry. In industry, CO is important in the production of many compounds, including drugs, fragrances, and fuels.

CO poisoning is the most common type of fatal air poisoning in many countries.



CARBON MONOXIDE

THE SILENT KILLER



500
deaths
per year

check yourself

do you know the signs of CO poisoning?



nausea



vomiting



shortness of breath



headache

leading causes

of carbon monoxide poisoning



furnace



stove



fireplace



water heaters



running engines



men are 2x more
likely to be affected
by CO poisoning
than women

NO_x

3. NO_x (Nitrogen oxides):

In atmospheric chemistry, NO_x is shorthand for nitric oxide (NO) and nitrogen dioxide (NO₂), the nitrogen oxides that are most relevant for air pollution.

These gases contribute to the formation of smog and acid rain, as well as affecting tropospheric ozone.

NO_x gases are usually produced from the reaction between nitrogen and oxygen during combustion of fuels, such as hydrocarbons, in air; especially at high temperatures.

NO_x (OXIDES OF NITROGEN)

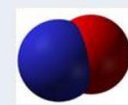


Nitric oxide|NO

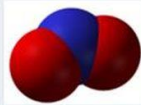
Nitrogen dioxide|NO₂

NO_x (Nitrogen Oxides)

Nitric Oxide
(NO)



Nitrogen Dioxide
(NO₂)



What are the Environmental Impacts of NO_x?

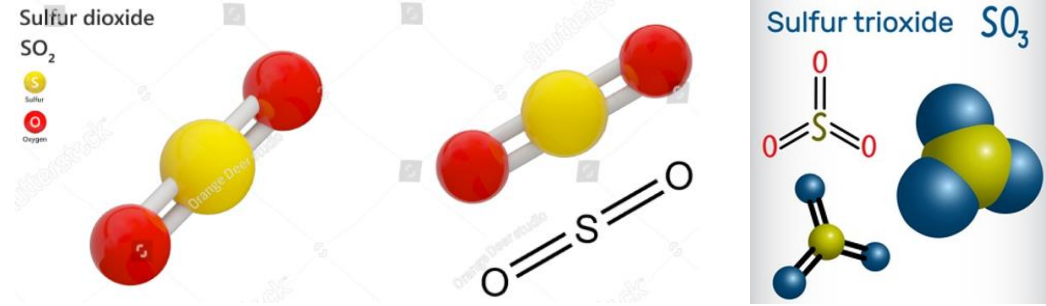


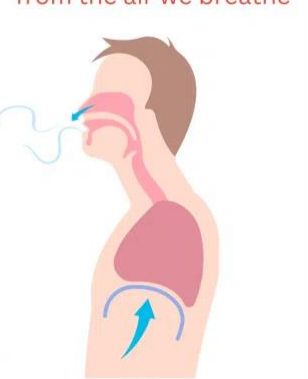
SO_x

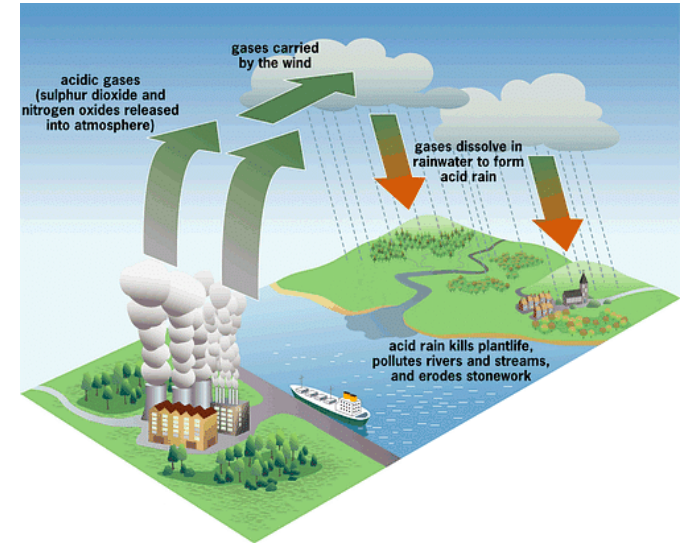
4. SO_x (Sulfur oxides = Sulphur oxides):

Sulfur oxides (SO_x) are a group of chemical compounds composed of sulfur and oxygen. The most common sulfur oxides are sulfur dioxide (SO₂) and sulfur trioxide (SO₃).

SO_x are pollutants that contribute to the formation of acid rain, as well as particulate pollution.



Sources	PATHWAY	HEALTH EFFECTS
Sulfur dioxide (SO₂) - Industry - Shipping 	Exposure to SO₂ comes from the air we breathe 	Impacts: - respiratory system - cardiovascular system Groups most at risk: - elderly - those with lung disease - children 



HC

5. HC (Hydrocarbons):

HC, any of a class of organic chemical compounds composed only of the elements carbon (C) and hydrogen (H).

The carbon atoms join together to form the framework of the compound, and the hydrogen atoms attach to them in many different configurations. Hydrocarbons are the principal constituents of petroleum and natural gas.

Ship exhaust gas contains HCs, which are unburnt fuel resulting from incomplete combustion.

HC can be present as methane (CH_4) when using liquefied natural gas (LNG).

Table 3 Typical composition of exhaust gases.

Component	N_2	O_2	CO_2	H_2O	NO_x	SO_x	CO	HC	Dust
%	77.50	13.75	6.25	0.025	0.212	0.17	0.005	0.005	0.0075



CH₄ (part of HC)

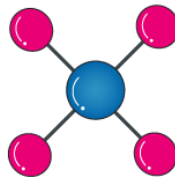
CH₄ (Methane):

Methane is a chemical compound with the chemical formula CH₄ (one carbon atom bonded to four hydrogen atoms). Methane is also a hydrocarbon.

It is the main constituent of natural gas. The abundance of methane on Earth makes it an economically attractive fuel, although capturing and storing it is difficult because it is a gas at standard temperature and pressure.

In the Earth's atmosphere, methane is transparent to visible light but absorbs infrared radiation, acting as a greenhouse gas.

METHANE STRUCTURE



Environmental Impact of Methane Emissions



Contribution to
Global Warming



Ecosystem
disruption



Impact on
vegetation and soil



Air quality and
health impacts



Water pollution



Explosive hazards

PM

6. PM (Particulate matter):

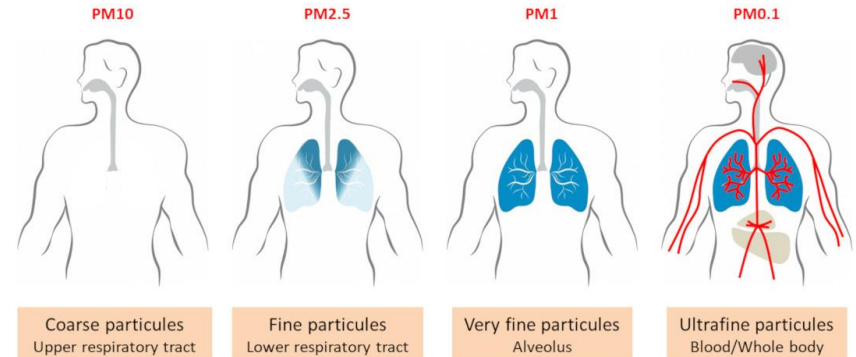
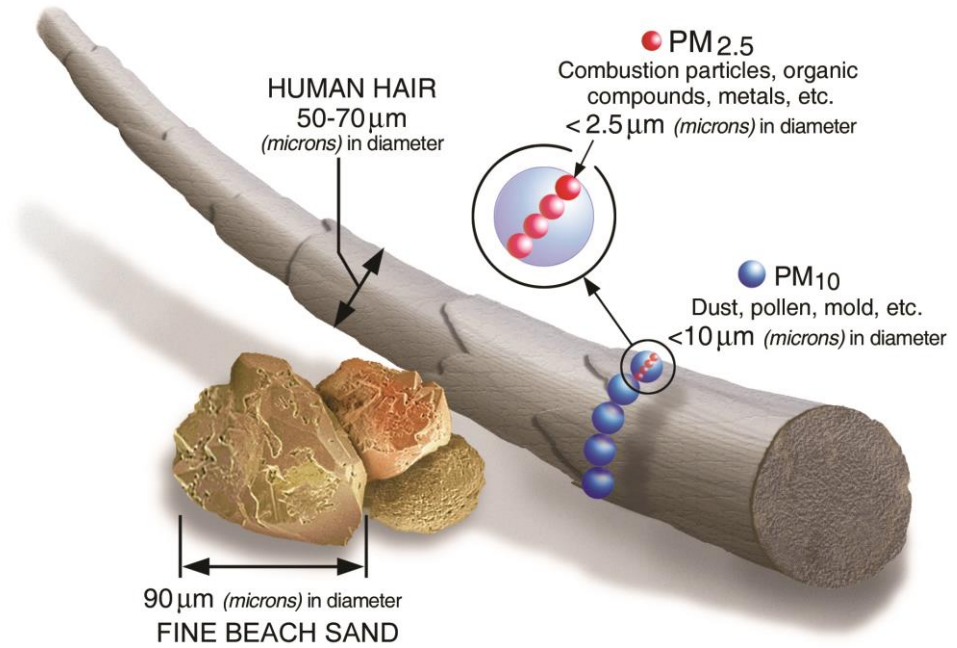
PM stands for particulate matter (also called particle pollution): the term for a mixture of solid particles and liquid droplets found in the air.

Some particles, such as dust, dirt, soot, or smoke, are large or dark enough to be seen with the naked eye.

Others are so small they can only be detected using an electron microscope.

PM₁₀: inhalable particles, with diameters that are generally 10 micrometers and smaller.

PM_{2.5}: fine inhalable particles, with diameters that are generally 2.5 micrometers and smaller.



BC



BC (Black carbon):

BC, commonly known as soot, is a component of fine particulate air pollution (PM_{2.5}).

It is formed by incomplete combustion of fossil fuels.

BC warms the atmosphere because it is very effective at absorbing light. It exacerbates warming of the air and surfaces in regions where it is concentrated, altering weather patterns and ecosystem cycles.

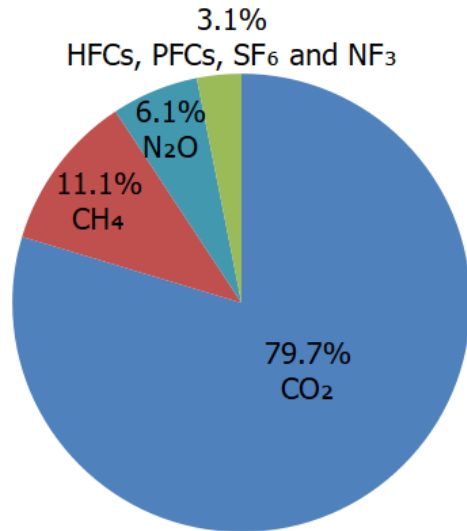
BC lasts only days to weeks in the atmosphere but has significant direct and indirect impacts on the climate, snow and ice, agriculture, and human health.



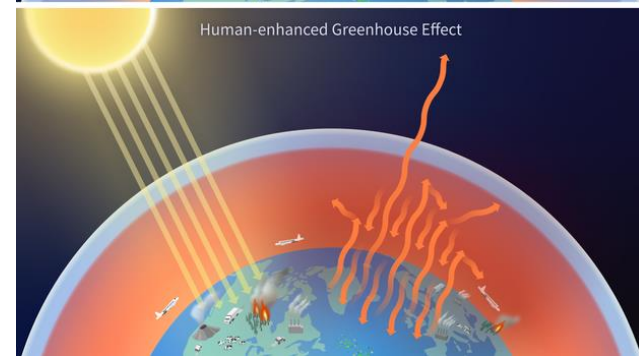
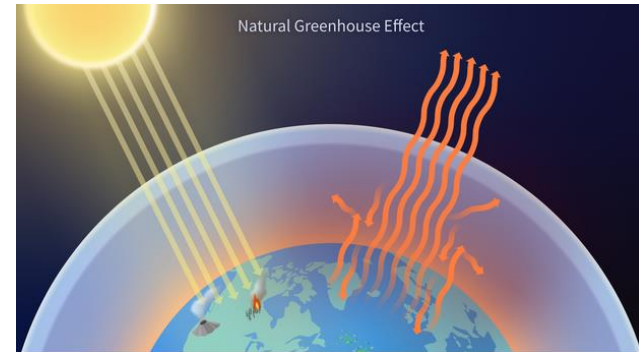
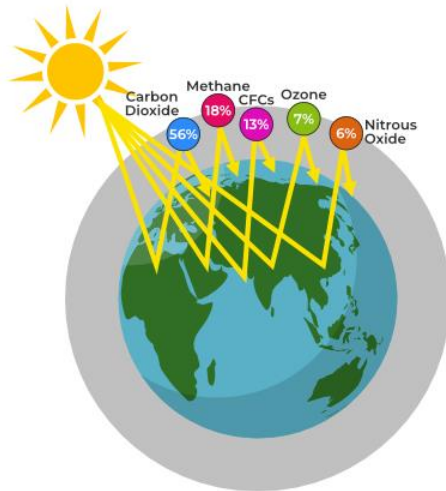
GHG

Greenhouse gases are gases in Earth's atmosphere that trap heat.

They let sunlight pass through the atmosphere, but they prevent the heat that the sunlight brings from leaving the atmosphere.

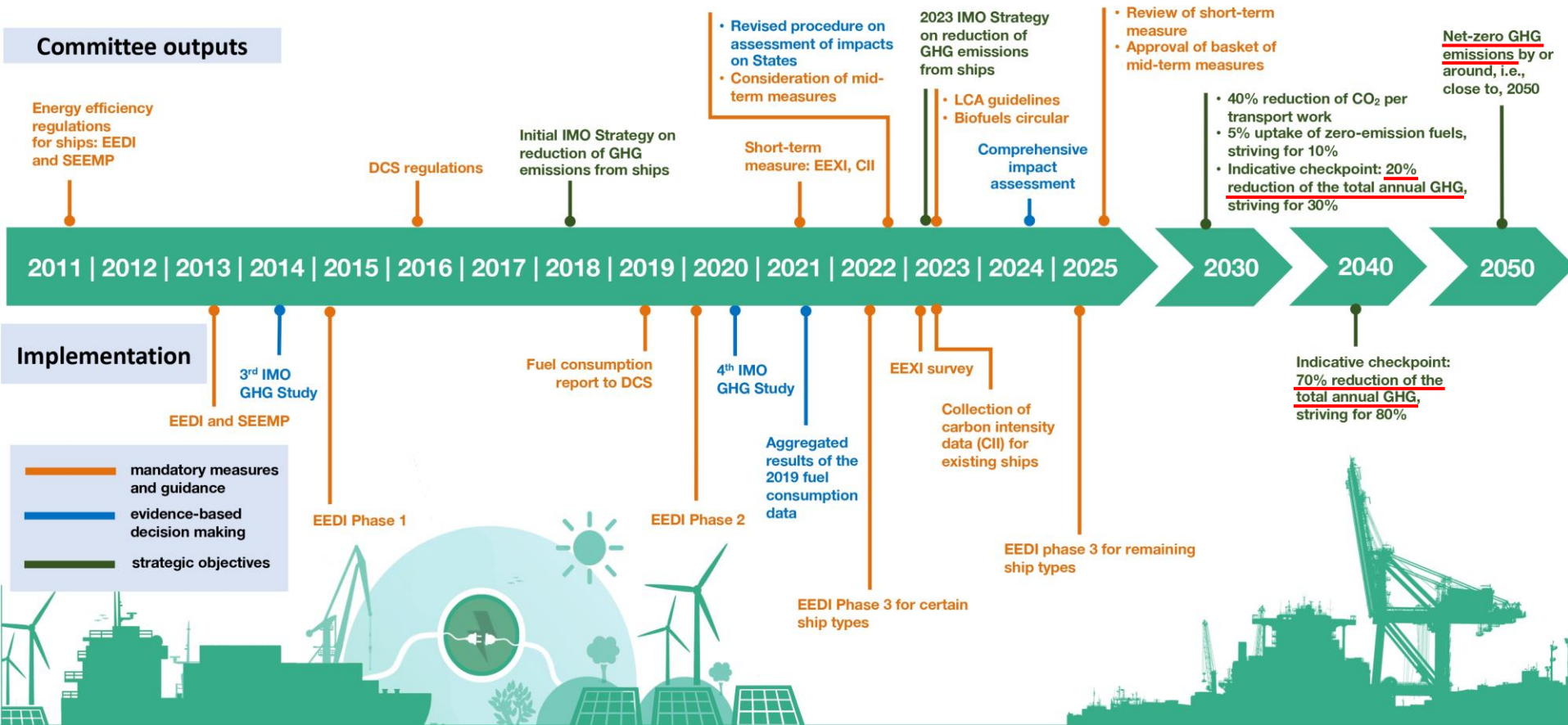


Greenhouse Gases



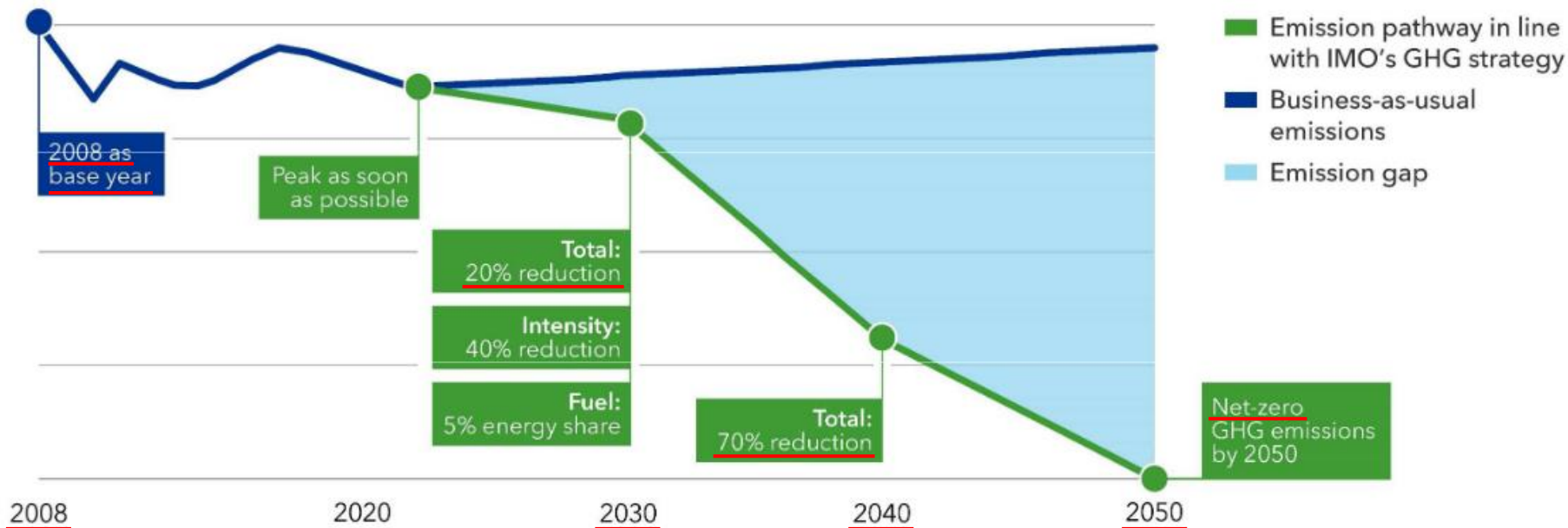
Addressing climate change

Over a decade of **regulatory action** to cut GHG emissions from shipping



Strengthened IMO strategy on GHG reductions

Units: GHG emissions



Total: Well-to-wake GHG emissions; **Intensity:** CO₂ emitted per transport work; **Fuel:** Uptake of zero or near-zero GHG technologies, fuels and/or energy sources

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IMO NOx regulation

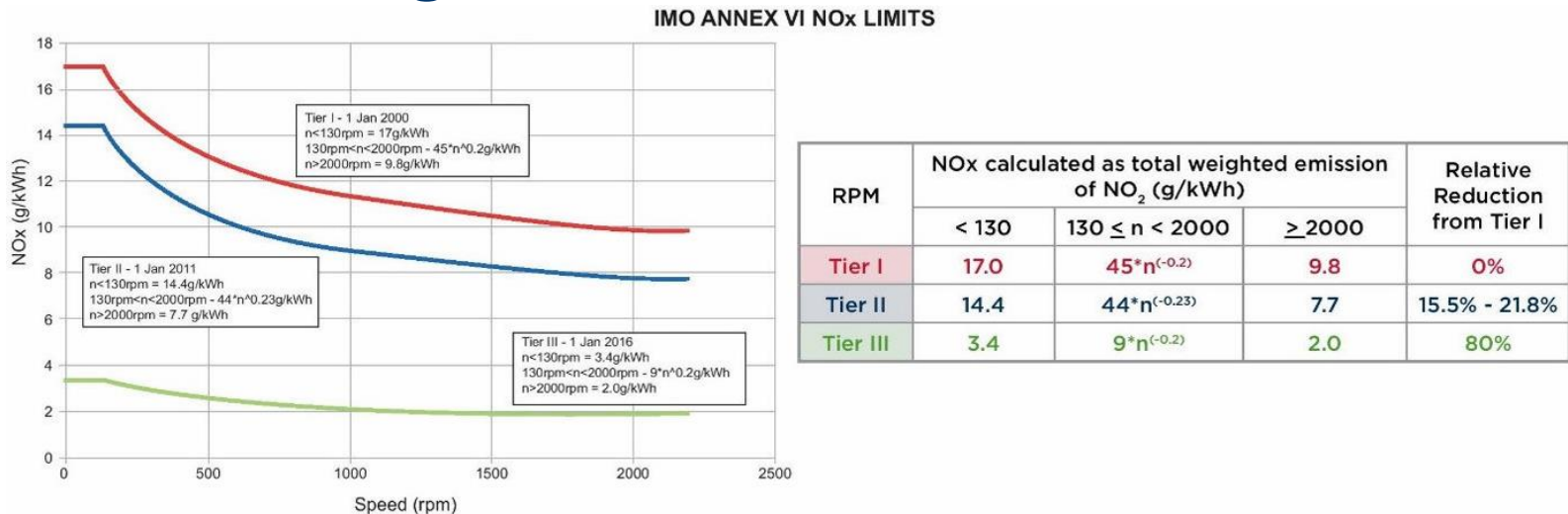


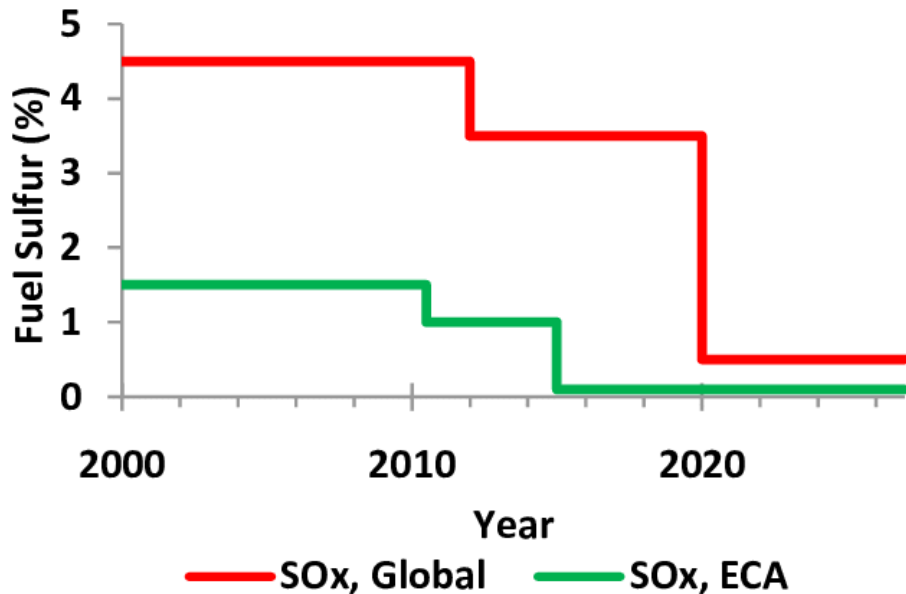
Figure 4. IMO MARPOL Annex VI Regulation 13 NO_x Limits, [15]

Tier I is compulsory for new diesel engines constructed or installed on ships from 01.01.2000.

Tier II is mandatory applicable for ships constructed after 01.01.2011.

Tier III compliance is required for ECA and 200 miles Exclusive Economic Zone of U.S. from 01.01.2016, as well as for Baltic and North Sea from 01.01.2021.

IMO SOx regulation



Tab. 2. Permissible sulphur content in fuel as defined in the MARPOL convention, Annex VI [5]

Date of limit application	Sulphur limits in fuel (%)	
	SECA	Global
Prior to 1 July 2010	1.5	4.5
On and after 1 July 2010	1.0	
2012		3.5
2015	0.1	
2020		0.5

ECA: Emission Control Area

SECA: SOx Emission Control Area

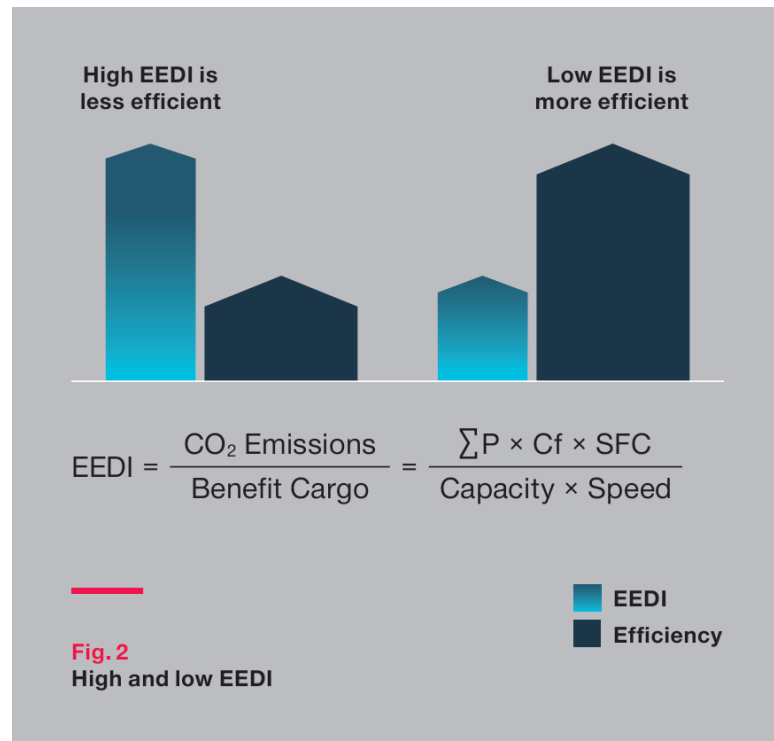
EEDI (Energy Efficiency Design Index)

The EEDI is used to calculate a vessel's energy efficiency. This is based on a complex formula, taking the ship's emissions, capacity, and speed into account.

The lower a ship's EEDI, the more energy-efficient it is and the lower its negative impact on the environment.

IMO regulations stipulate that ships must meet a minimum energy efficiency requirement, so their EEDI must not exceed a given threshold.

Ships commissioned after January 1, 2013 and weighing 400 GT or more have to meet the requirements.



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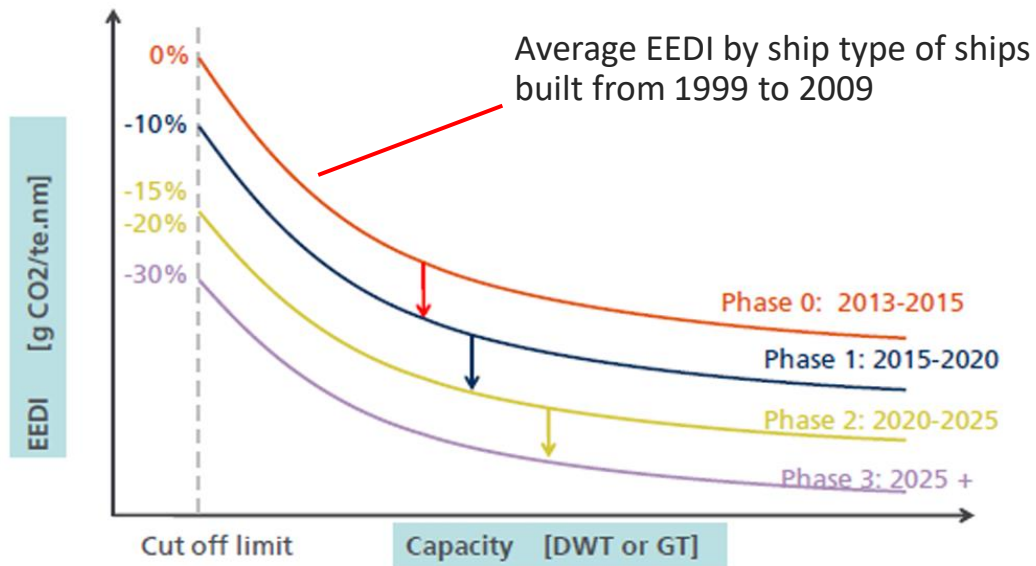


Figure 55: Phases for reduction factors of EEDI (adopted from [59])

EEDI (Energy Efficiency Design Index)

Efficiency-improving devices



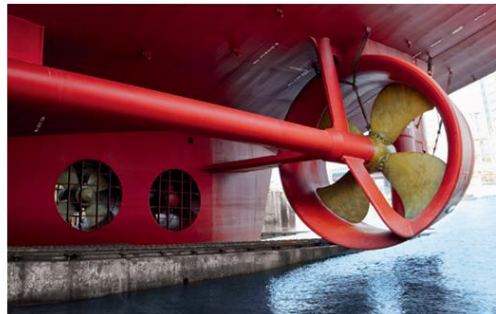
MAN Alpha EcoBulbs

Efficiency improvements of up to 6 % (proven in model tests).



Customized Kappel propeller designs

Kappel propellers improve propulsion efficiency by up to 6 % while reducing power consumption, emissions and noise. This enable vessels to reduce EEDI/EEOI values and achieve higher 'energy classes'. Kappel propellers are suitable for newbuilds and retrofits.



Alpha High Thrust (AHT) nozzles

AHT nozzles optimize the propeller thrust and pulling performance of heavy duty vessels. The individually customized designs increase bollard pull and limit free-sailing resistance.

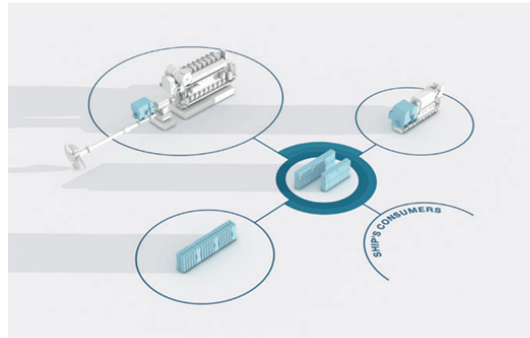
EEDI (Energy Efficiency Design Index)

Efficiency-improving technologies



CPP EcoOptimizer for ~~CPP~~ systems

The EcoOptimizer combines the Alpha-tronic 3000 propulsion control system with individual main engine SFOC maps and MAN Alpha Controllable Pitch Propellers to enable fuel savings of up to 6 %.



MAN HyProp ECO and MAN HyProp Battery

MAN HyProp ECO is a flexible hybrid propulsion system for controlling the power delivered by or to the shaft machine in the most efficient way. It overcomes the constraint on constant

speed propulsion machinery by utilizing variable speed drive (VSD) technology at the shaft generator/motor. MAN HyProp ECO can reduce fuel oil consumption by 10 – 15 %.



EEDI (Energy Efficiency Design Index)

The EEDI assesses the energy consumption of a vessel under normal seafaring conditions, taking into account the energy required for propulsion and the hotel load for the crew.

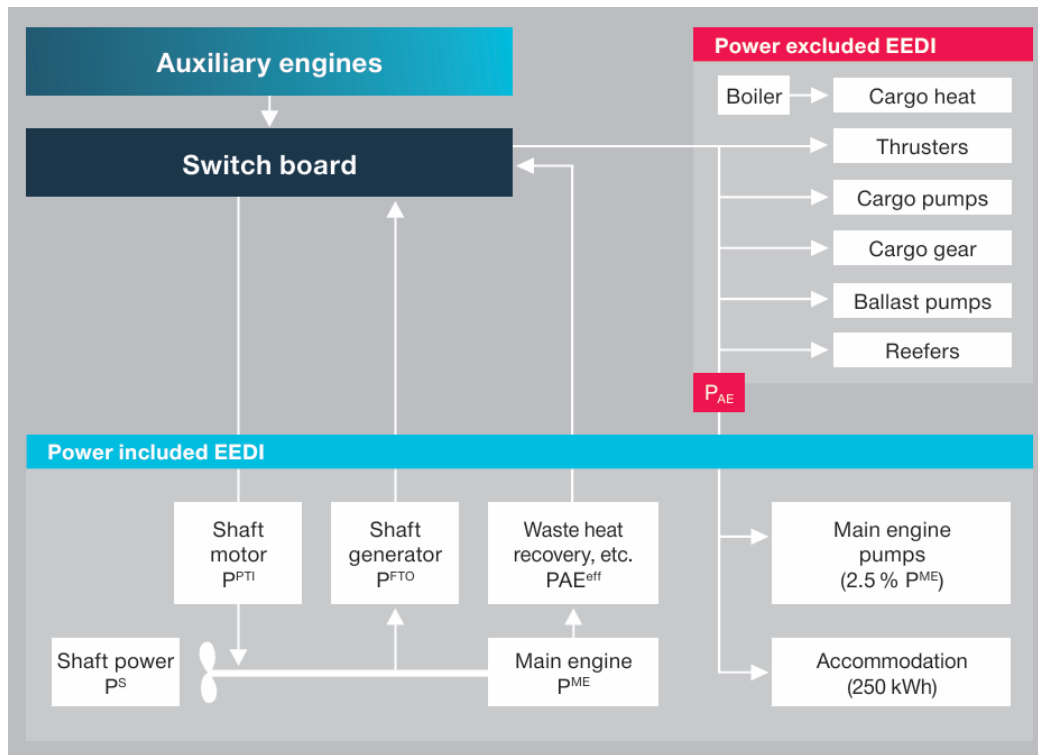


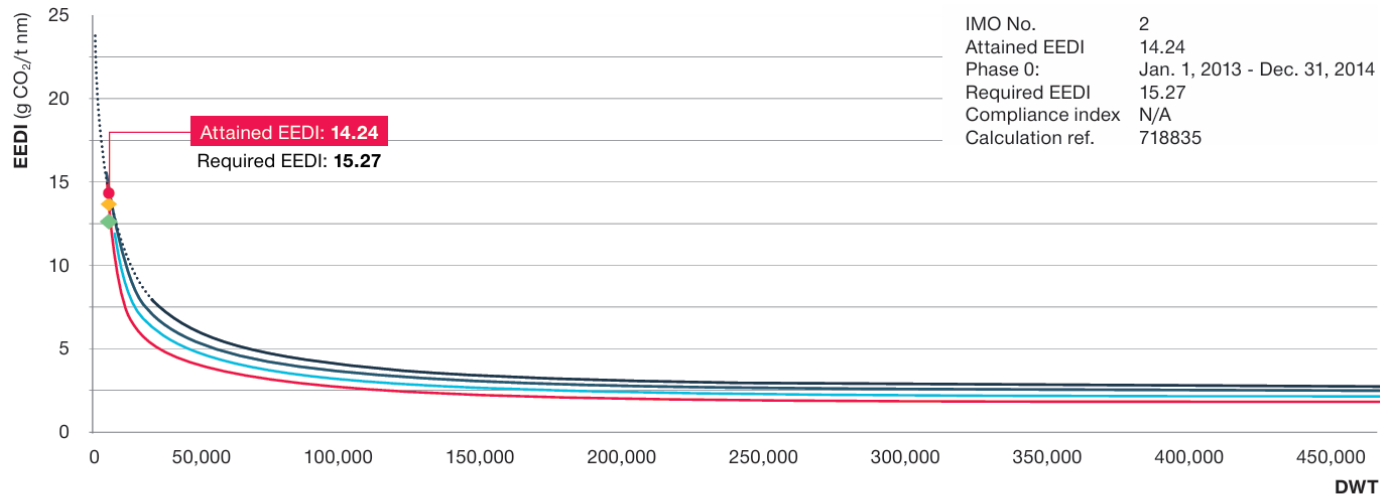
Fig. 5

Flow of energy considered by EEDI



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EEDI (Energy Efficiency Design Index)



Requirement tanker (2008)

- DWT design draft
7,900 t
- Main engine
3,360 kW (MAN 6L32/44CR)
- Auxiliary engine
1 x 1,290 kW (MAN 6L21/31)
- Generator efficiency
93 %
- Speed
13.3 knots
- Fuels
Diesel/gas oil, ISO 8217, DMC – DMX

- New ships from 1/1/2013
- New ships from 1/1/2015
- New ships from 1/1/2020
- New ships from 1/1/2025
- ⋮ Reference line

Sensitivity

- Attained EEDI: 14.24
- ◆ Attained EEDI: 13.84
(with engine MAN 9L27/38; 3,060 kW and 185 g/kWh SFC*)
- ◆ Attained EEDI: 12.64
(using MAN 6L35/44DF)

* 85 % MCR

Assumptions and considerations:

All variations are only achieved by changing the main engine characteristics.

Fig. 10
Requirement tanker

$$\left[\prod_{j=1}^M f_j \right] \left[\sum_{i=1}^{AME} P_{ME(i)} \cdot SFC_{ME(i)} \cdot C_{FME} \right] \cdot (P_{AE} \cdot C_{FAE} \cdot SFC_{AE}) \cdot \left[\left[\prod_{j=1}^M f_j \cdot \sum_{i=1}^{PPI} P_{PT(i)} - \sum_{i=1}^{noff} f_{off(i)} \cdot P_{AE(i)} \right] C_{FAE} \cdot SFC_{AE} \right] \cdot \left[\sum_{i=1}^{noff} f_{off(i)} \cdot P_{AE(i)} \cdot C_{FME} \cdot SFC_{ME} \right]$$

$$f_i \cdot f_c \cdot f_i \cdot \text{Capacity} \cdot V_{ref} \cdot f_w$$

Transport work

Engine power (P)

Individual engine power depending on application (e. g. $P_{ME} = 75\%$ maximum continuous rating for diesel-mechanic propulsion)

$P_{eff(i)}$ Main engine power reduction due to individual technologies for mechanical energy efficiency

$P_{AEff(i)}$ Auxiliary engine power reduction due to individual technologies for electrical energy efficiency

$P_{PT(i)}$ 75 % of rated power consumption of shaft motor

P_{AE} Combined installed power of auxiliary engines

$P_{ME(i)}$ Individual power of main engines

CO₂ emissions (C)

CO₂ emission factor based on type of fuel used by given engine

C_{FME} Main engine composite fuel factor

C_{FAE} Auxiliary engine fuel factor

$C_{FME(i)}$ Main engine individual fuel factors

Ship design parameters

V_{ref} Ship speed at reference conditions (see P_{ME} definition, etc.)

$Capacity$ Deadweight tonnage (DWT) rating for bulk ships and tankers; a percentage of DWT for container ships; DWT indicates how much can be loaded onto a ship; gross tonnage for passenger ships (cruise)

Specific fuel consumption (SFC)

Fuel use per unit of engine power

SFC_{ME} Main engine (composite)

SFC_{AE} Auxiliary engine

SFC_{AE*} Auxiliary engine (adjusted for shaft generators)

$SFC_{ME(i)}$ Main engine (individual)

Correction and adjustment factors (F)

Non-dimensional factors that were added to the EEDI equation to account for specific existing or anticipated conditions that would otherwise skew the ratings of individual ships

$f_{off(i)}$ Availability factor of individual energy efficiency technologies (=1.0 if readily available)

f_i Correction factor for ship-specific design elements, e.g. ice-classed ships which require extra weight for thicker hulls

f_w Coefficient indicating the decrease in ship speed due to weather and environmental conditions

f_i Capacity adjustment factor for any technical/regulatory limitation on capacity (=1.0 if none)

f_c Cubic capacity correction factor (for chemical tankers, LNG carriers and RoPax)

f_i Correction factor to compensate deadweight losses through cargo-related equipment like cranes, RoRo ramps, etc.



SEEMP (Ship Energy Efficiency Management Plan)

SEEMP is a structured, practical tool for helping shipowners to manage their environmental performance and improve operational efficiency.

The SEEMP consists of three parts:

Part I: Ship management plan to improve energy efficiency

Part II: Ship fuel oil consumption data collection plan

Part III: Ship operational carbon intensity plan

Part I of the SEEMP is mandatory and must be kept on board all ships above 400 GT.

A verified Part II is mandatory for all ships above 5,000 GT as part of the Data Collection System.

A verified Part III is required for all ships subject to the Carbon Intensity Indicator (CII) – i.e. cargo, Ro-Pax and cruise passenger vessels above 5,000 GT.



SEEMP (Ship Energy Efficiency Management Plan)

The SEEMP requires owners and operators to consider and implement new technologies and practices when seeking to optimize the performance of the ship. Typical examples of improving operational efficiency and carbon intensity include:

- Speed optimization
- Weather routing
- Hull monitoring and maintenance
- Installation of heat recovery systems

SEEMP Parts I and III are generally divided into the following sections:

- Goal
- Planning and implementation of measures
- Monitoring
- Self-evaluation/improvement

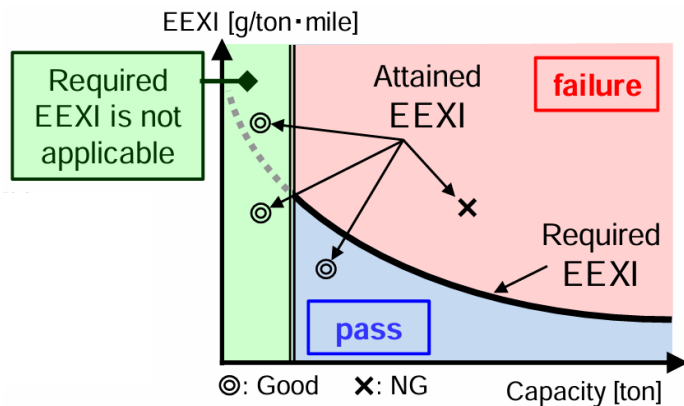
EEXI (Energy Efficiency eXisting ship Index)

From 1 January 2023 it became mandatory for all ships to calculate their attained EEXI to measure their energy efficiency regardless of ship's delivery date.

It must be calculated for ships of 400 gt and above, in accordance with the different values set for ship types and size categories.

The EEXI is a measure related to the technical design of a ship.

In case no changes to the energy efficiency are applied, ships have to attain EEXI approval only once in a lifetime. The initial certification was surveyed during the first periodical survey in 2023.



EEXI (Energy Efficiency eXisting ship Index)

- EEXI is calculated by the same formula as EEDI.

EEXI [g/ton • mile]=

$$\frac{\left(\prod_{j=1}^M f_j \right) \left(\sum_{i=1}^{nME} P_{ME(i)} \cdot C_{FME(i)} \cdot SFC_{ME(i)} \right) + (P_{AE} \cdot C_{FAE} \cdot SFC_{AE}) + \left\{ \left(\prod_{j=1}^M f_j \cdot \sum_{i=1}^{nPTI} P_{PTI(i)} - \sum_{i=1}^{neff} f_{eff(i)} \cdot P_{AEff(i)} \right) \cdot C_{FAE} \cdot SFC_{AE} \right\} - \left(\sum_{i=1}^{neff} f_{eff(i)} \cdot P_{eff(i)} \cdot C_{FME} \cdot SFC_{ME} \right)}{f \cdot f_c \cdot f_i \cdot Capacity \cdot V_{ref}}$$



Concept formula

$$\text{EEXI [g/ton • mile]} = \frac{\text{CO}_2 \text{ Conversion factor} \times \text{SFC [g/kW • h]} \times \text{Engine Power [kW]}}{\text{Capacity [ton]} \times \text{EEXI Speed [knots]}}$$



CO₂ emissions (gram) from a ship when ship sail transport 1 (ton) cargo for 1 (nautical mile)

CO ₂ Conversion factor (C _F)	C _F corresponds to the fuel used when determining SFC (DM grade: 3.206)
SFC	Fuel consumption at 75%MCR (M/E), at 50%MCR (A/E)
Engine Power	75% of the rated installed power (MCR) (In case of EPL, 83%MCR _{lim})
Capacity	Deadweight (For containerhips, 70% of the deadweight)
EEXI Speed (V _{ref})	Ship speed at 75%MCR under the draught condition corresponding to the capacity



EEXI (Energy Efficiency eXisting ship Index)

- The formula of EEXI is the same as EEDI, but some parameters' definitions are different.

Concept formula

$$\text{EEXI [g/ton} \cdot \text{mile]} = \frac{\text{CO2 Conversion factor} \times \text{SFC [g/kW} \cdot \text{h]} \times \text{Engine Power [kW]}}{\text{Capacity [ton]} \times \text{EEXI Speed (Vref) [knot]}}$$

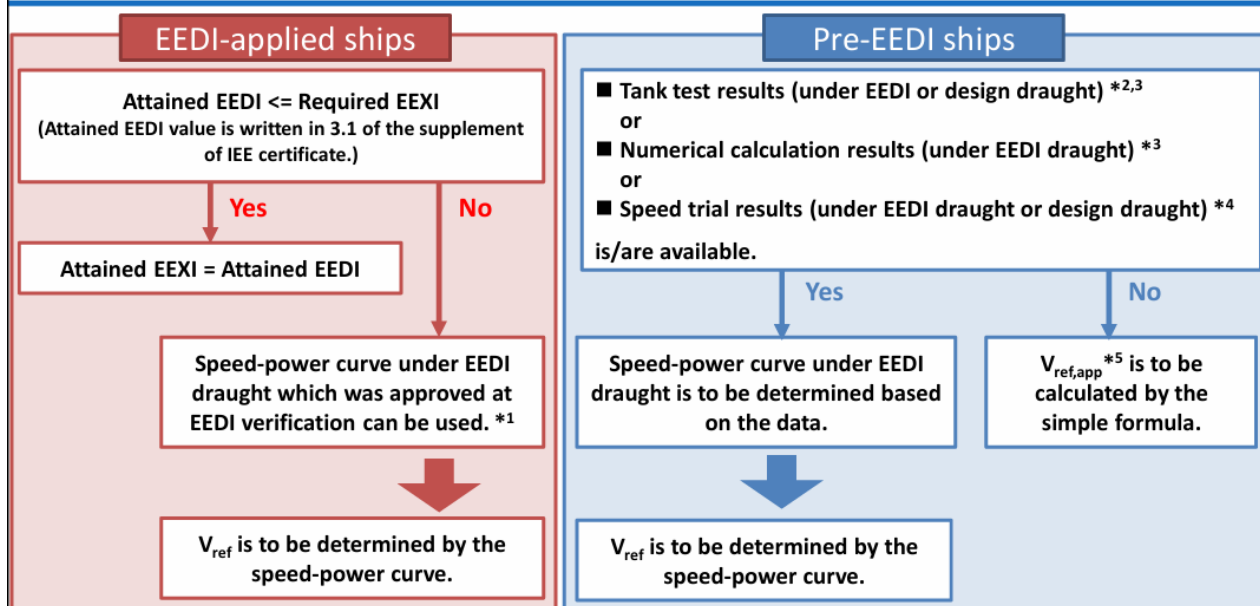
SFC	Fuel consumption at 75%MCR (M/E), at 50%MCR (A/E) specified in NOx technical file • In cases where the installed engines don't have NOx technical file, approximated default values including margin, SFC_{app} (i.e. M/E: 190 g/kW · h, A/E: 215 g/kW · h), are available. • In cases where the NOx regulation doesn't apply to the propulsion system (e.g. steam turbine, etc.), SFC specified by the manufacturer or confirmed by the verifier is available.
Engine Power (P_{ME})	75% of the rated installed power (MCR) (In cases where the propulsion system is diesel electric or steam turbine, P_{ME} is 83% of MPP or MCR.) • In cases where EPL is installed, P_{ME} is 83% of the limited installed power (MCR_{lim}).
V_{ref}	Ship speed at P_{ME} and under the maximum summer load draught (for container ships, under 70%DWT draught.) • In cases where both of tank test results and speed trial results are not available, an approximated ship speed including margin, $V_{\text{ref,app}}$ is to be calculated by the simple formula. The parameters of this formula are ship type, DWT, and MCR.



EEXI (Energy Efficiency eXisting ship Index)

Methods for obtaining V_{ref}

ClassNK



*1 The approved speed-power curve is available without any corrections.

*2 The tank test results can be corrected/calibrated by numerical calculation such as CFD, etc.

*3 In case of using numerical calculations, estimation process and methodology of the power curves are to be submitted. (It should include documentation on consistency with the defined quality standards and the verification of the numerical setup with parent hull or the reference set of comparable ships.)

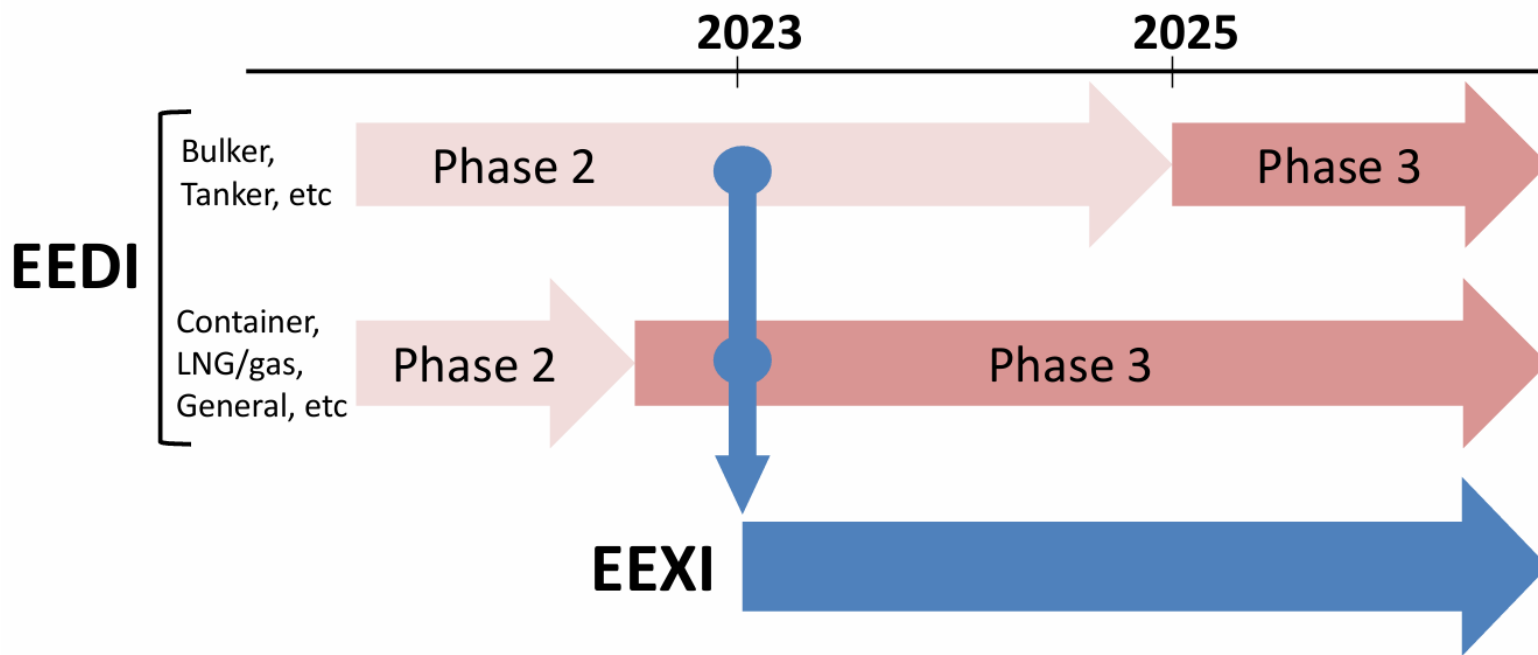
*4 The sea conditions and ship speed should have been measured in accordance with ISO 15016:2002 or the equivalent and the measured ship speed was calibrated, if necessary, by taking into account the effects of wind, tide, waves, etc. If the speed trial was carried out under design draught, the ship speed shall be calibrated under EEDI draught by using Admiralty Coefficient, etc.

*5 $V_{ref,app}$ is an approximated ship speed obtained by a certain correction applies to the average ship speed of each ship type and size (including margin).



EEXI (Energy Efficiency eXisting ship Index)

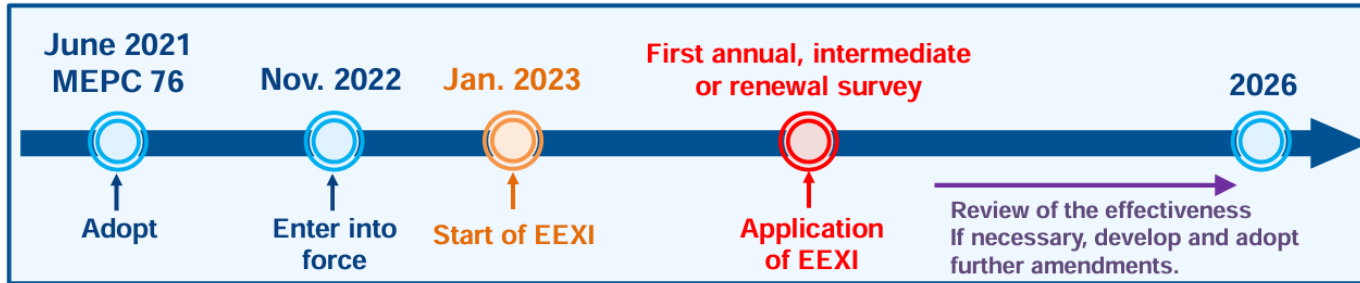
Almost the same level as **required EEDI for new ships as of 2023***



EEXI (Energy Efficiency eXisting ship Index)

Timeline of EEXI regulation

ClassNK



■ Timing of EEXI application

- ✓ The amendments to MARPOL ANNEX VI will enter into force on 1 November 2022.
- ✓ EEXI requirements will start from **1st January 2023**.
- ✓ The EEXI verification shall take place at the following timing.

Ships delivered **before**
1 January 2023

First annual, intermediate or renewal survey of the International Air Pollution Certificate (IAPP Certificate), whichever is the first, **on or after 1 January 2023**

Ships delivered **on or after**
1 January 2023

Initial survey of the International Energy Efficiency Certificate (IEE Certificate)

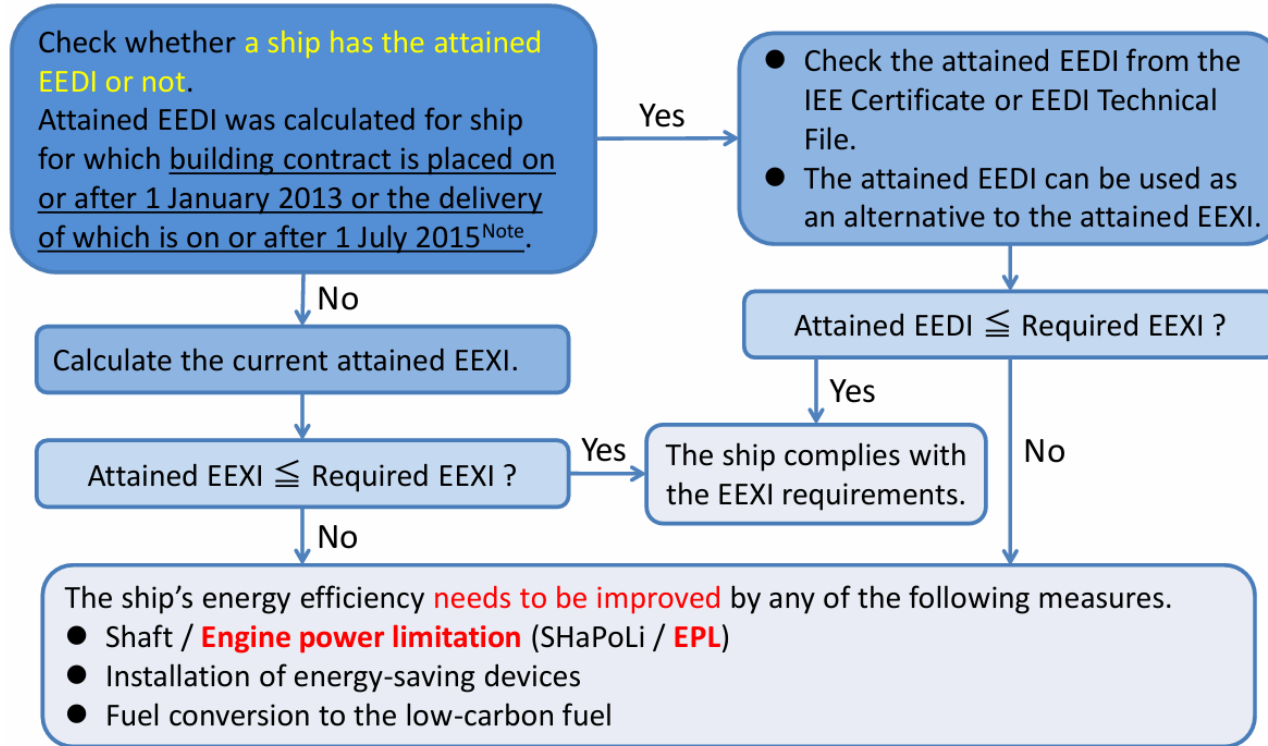


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EEXI (Energy Efficiency eXisting ship Index)

Flow chart of EEXI application

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Note : In case of LNG carrier and Cruise passenger ship, building contract is placed on or after 1 September 2015 or the delivery of which is on or after 1 September 2019.



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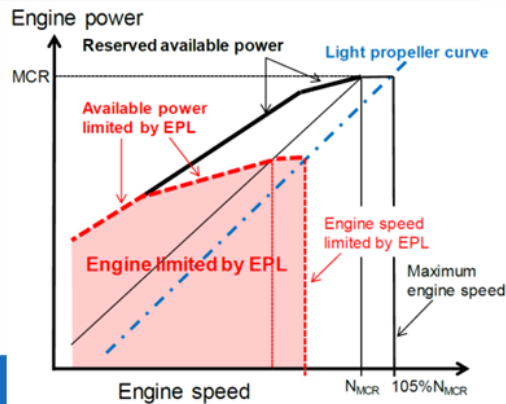
EEXI (Energy Efficiency eXisting ship Index)

■ What is Engine Power Limitation (EPL) ?

- ✓ Engine Power Limitation (EPL) is a system to improve a ship's energy efficiency by limiting the ship's engine power within the optimum engine setting. As a result, the ship speed will be limited.
- ✓ EPL consists of a simple device which can easily limit the maximum engine power by **adjusting a fuel index limiter** on the engine control system without retrofitting a complicated system within the current regulatory framework.
- ✓ EPL can be easily installed in a short time during a port without updating EIAPP certificate and the NOx technical file.
- ✓ EPL can be released in the adverse weather conditions. Therefore, the limited engine power does not have to meet the minimum power requirement.



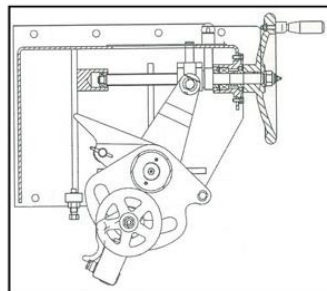
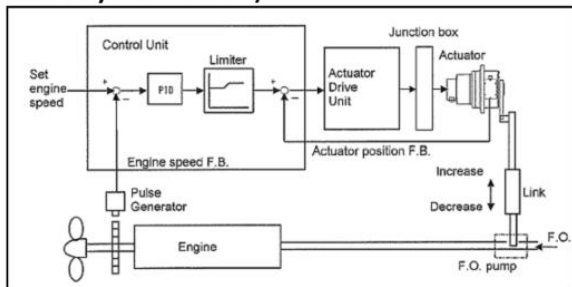
EPL can be utilized as **one of the effective measures** to improve energy efficiency of existing ships in terms of EEXI.



EEXI (Energy Efficiency eXisting ship Index)

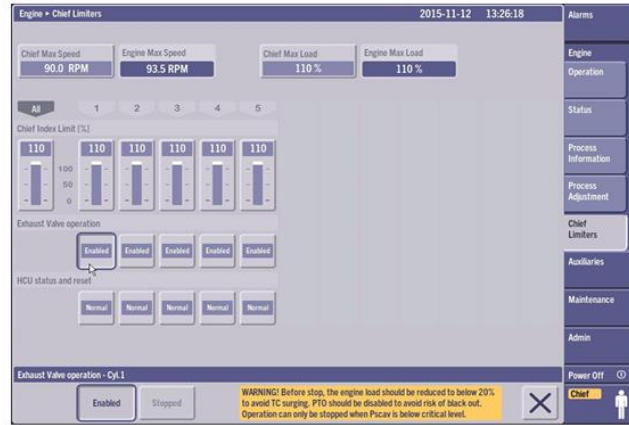
■ Mechanically driven type engine such as old engine type

1. Changing a set of governor's fuel index limiter
2. Adjusting the Mechanical Stop Screw
3. Sealing the Mechanical Stop Screw by wire and so on (confirmed at statutory survey and PSC)



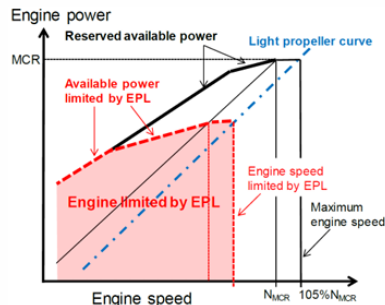
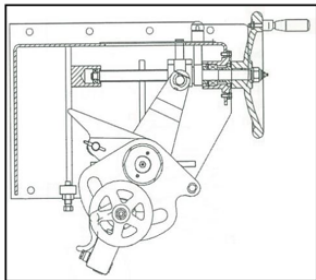
- Electronically controlled type engine such as new engine type

- (In case of MAN B&W engines, setting the fuel index limit in Chief Limiters)



Since the electronically controlled type engine is not physically sealed unlike the mechanical driven type engine, it is confirmed that the EPL had not been released without permission since the last confirmation by **checking the data recorded in the data logging program.**

EEXI (Energy Efficiency eXisting ship Index)



■ Principal requirements

- The available power is to be limited by locking fuel index.
- EPL cannot be released without permission from the ship master or the officer in charge of navigational watch (OICNW).
- If EPL is un-limited due to the purpose of securing the safety of a ship or saving life at sea, the reason and relevant information are to be recorded in Onboard Management Manual (OMM).
- EPL system (or each sub system) should be tamper-proof.
- EPL system should be accompanied by OMM for EPL that should be on board the ship for inspection.

■ Contents of OMM

Original MCR (kW x rpm), MCR after installing EPL: MCR_{lim} (kW x rpm), Technical description of EPL system, Sealing method (mechanically controlled engine), Locking and monitoring method (electronically controlled engine), Procedures and methods for releasing EPL, Time required for un-limiting EPL, Procedures for survey by the Administration/RO, Procedure for the report on release of EPL, Administrator of the EPL system, etc.



EEXI (Energy Efficiency eXisting ship Index)

The un-limiting EPL is only allowed for the purpose of securing the safety of a ship or saving life at sea, consistent with regulation 3.1 of MARPOL Annex VI.

Examples)

- Operating in adverse weather and ice-infested waters, or avoidance voyaging in such areas
- Participation in search and rescue operations
- Avoidance of pirates
- Engine maintenance (e.g. removing soot, etc.)

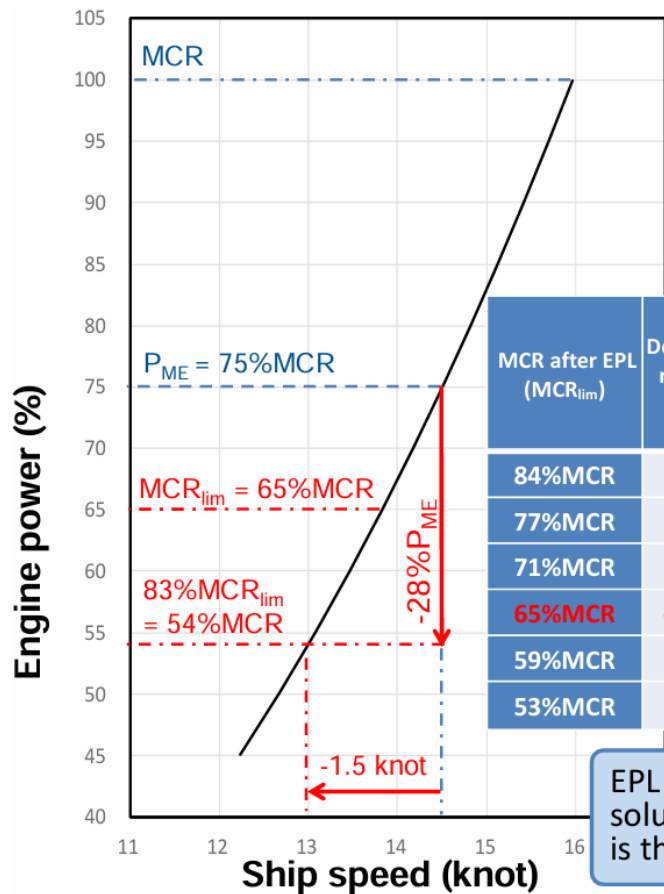


■ Necessary procedures in cases where EPL is un-limited

- Recording the status in OMM (e.g. reason of the un-limiting, ship speed, maximum un-limited power, beaufort number and wave height, position and timestamp, etc.)
- Notifying Administration or RO
- Reactivating/Replacing EPL system immediately after the risks have been prevented
- Confirmation of the reactivated/replaced EPL system by Administration or RO (Remote confirmation may be acceptable.)



EEXI (Energy Efficiency eXisting ship Index)



EEXI after EPL [g/ton • mile] =

Conversion Factor × SFC × **Engine Power (83% MCR_{lim})**

Capacity × **Ship Speed at 83% MCR_{lim}**

$$\left(\begin{array}{l} \text{Power} \propto \text{Speed}^3 \Rightarrow \text{EEXI} \propto \text{Speed}^2 \\ \text{Speed} \propto \text{Power}^{1/3} \Rightarrow \text{EEXI} \propto \text{Power}^{2/3} \end{array} \right)$$

MCR after EPL (MCR _{lim})	Decrease rate of P _{ME}	Decrease rate of Speed	Original Speed (V _{ref} at 75%MCR) [knot]	Speed after EPL (V _{ref} at 83%MCR _{lim}) [knot]	Decrease of Speed (ΔV) [knot]	Improvement rate of EEXI
84%MCR	-7%	-3%	14.5	14.1	-0.4	+5%
77%MCR	-15%	-5%	14.5	13.8	-0.7	+10%
71%MCR	-22%	-8%	14.5	13.4	-1.1	+15%
65%MCR	-28%	-11%	14.5	13.0	-1.5	+20%
59%MCR	-35%	-13%	14.5	12.6	-1.9	+25%
53%MCR	-41%	-16%	14.5	12.1	-2.4	+30%

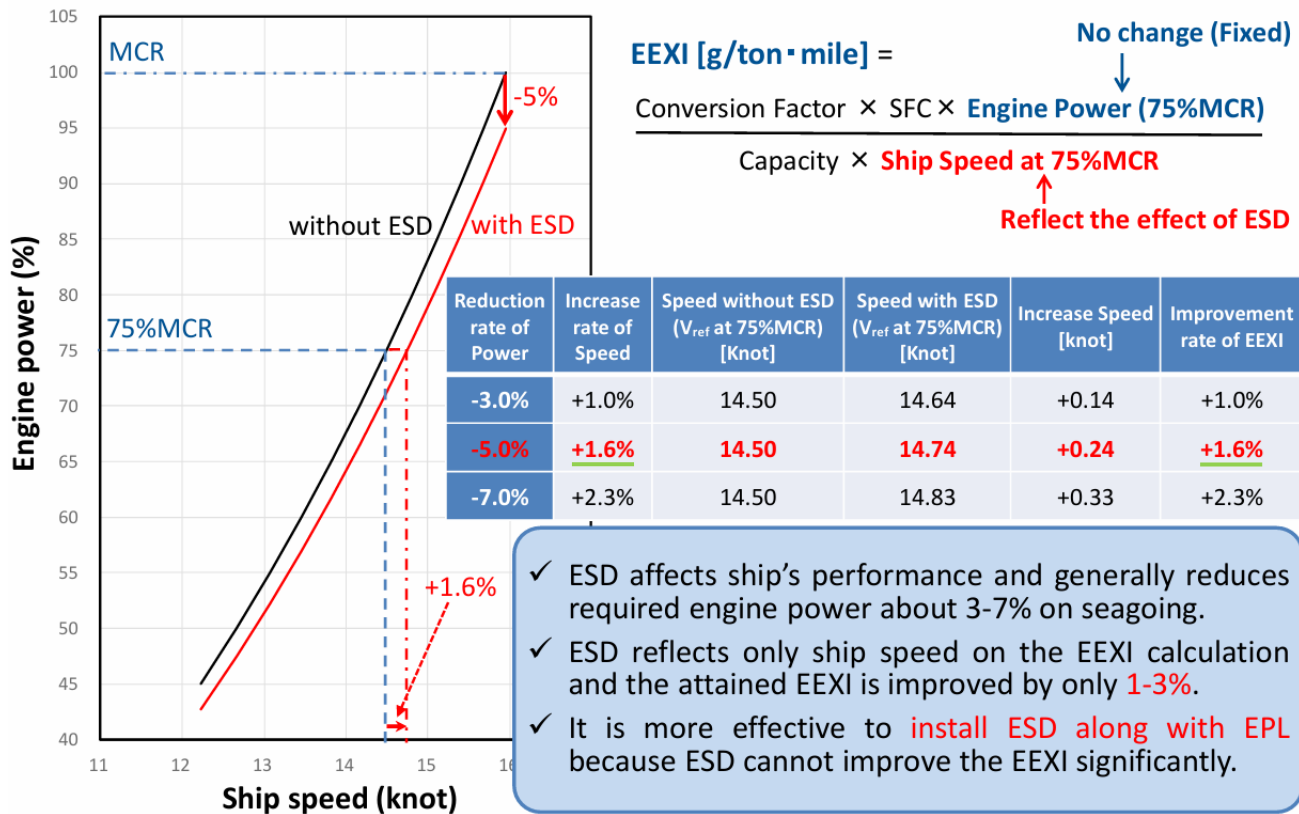
$$\frac{0.72}{0.89} = 0.809$$

EPL is a highly effective measure and a cost-effective solution to improve the ship's attained EEXI. Thus, EPL is the **most effective measure** for EEXI improvement.



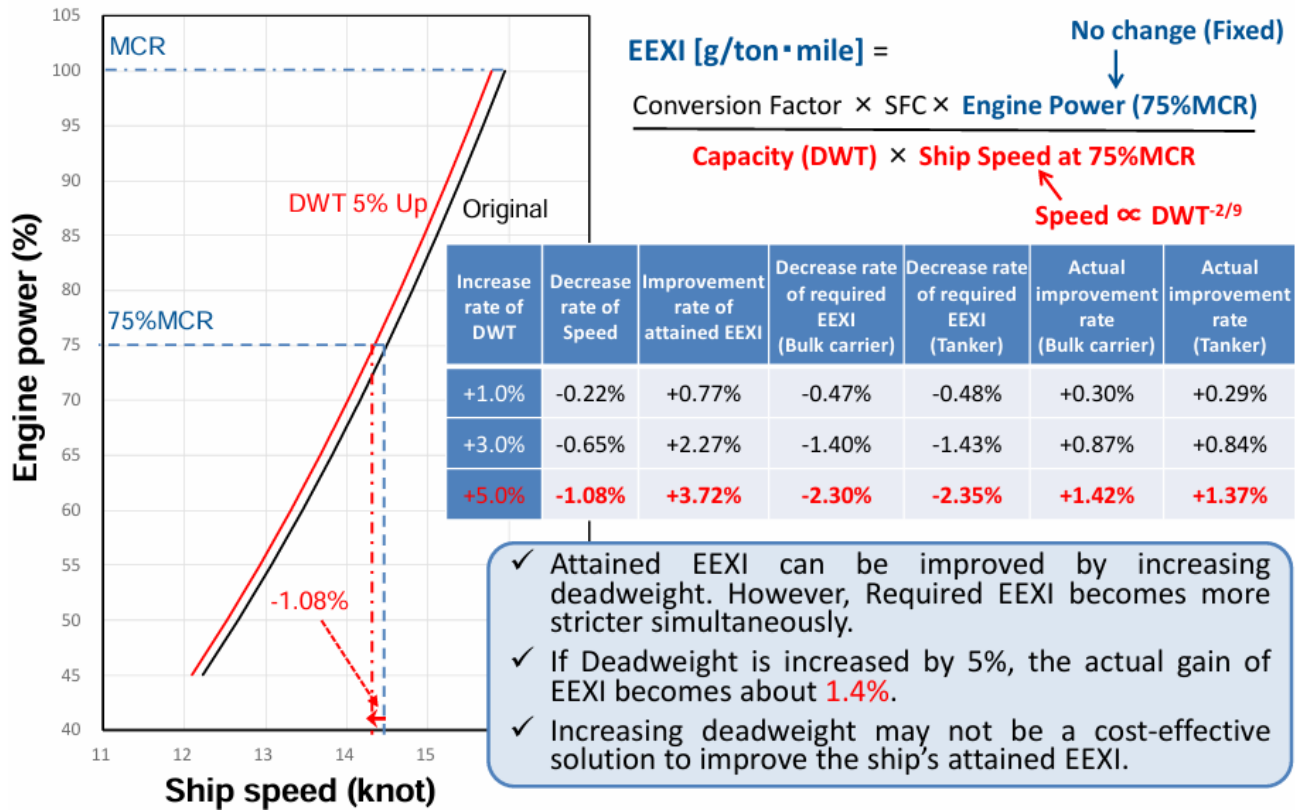
EEXI (Energy Efficiency eXisting ship Index)

Example of improvement of EEXI by Energy Saving Device (ESD) **ClassNK**



EEXI (Energy Efficiency eXisting ship Index)

Example of improvement of EEXI by increasing deadweight **ClassNK**

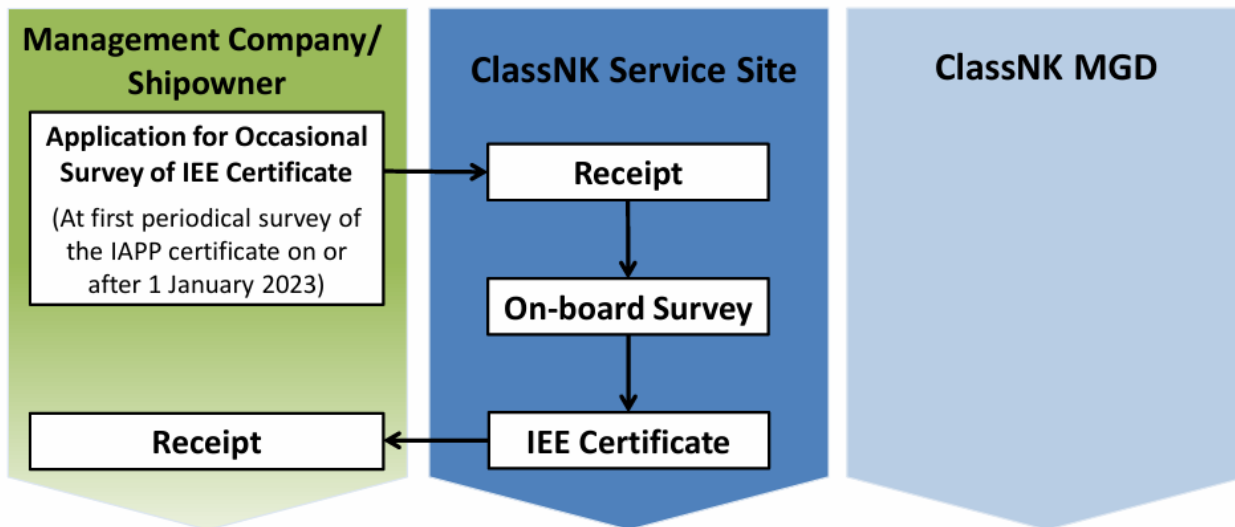


EEXI (Energy Efficiency eXisting ship Index)

■ Flow of EEXI Verification Process

1. Achieve their Required EEXI by their Attained EEDI
2. Other cases

1. Achieve their Required EEXI by their Attained EEDI

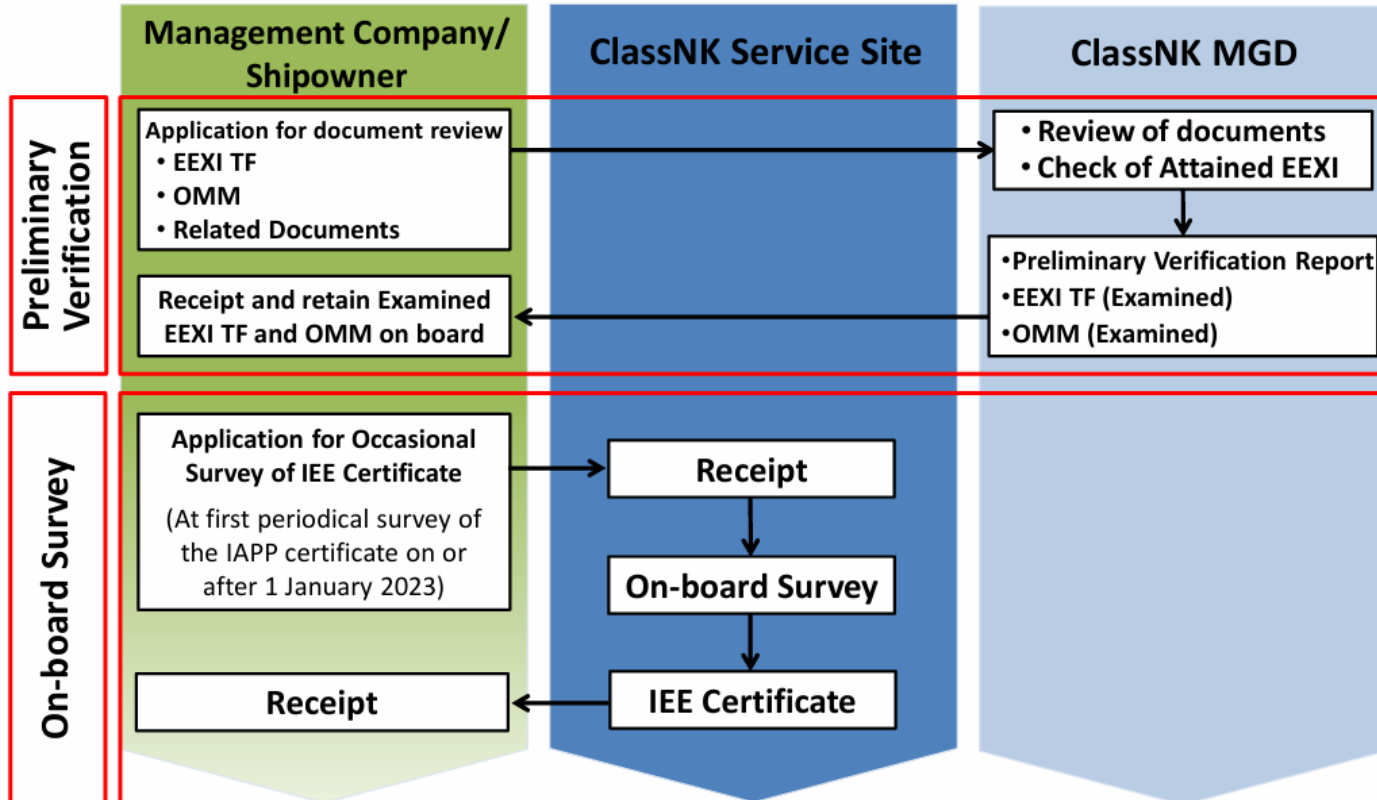


Note : The attained EEDI and the EEDI technical file of the ship can be used as an alternative to the attained EEXI and the EEXI technical file, so there is no need to prepare the EEXI technical file in advance.



EEXI (Energy Efficiency eXisting ship Index)

2. Other cases



EEXI (Energy Efficiency eXisting ship Index)

Points of attention for EEXI (1/2)

ClassNK

EEXI Calculation

- Documents on **V_{ref}** and **SFC** are required for EEXI calculation.
- EEXI value can be conservatively calculated by using the ship speed given by simple formula and the default value of SFC. However, **calculation of accurate EEXI** counts to minimize the ship's operation.
- Calculation of accurate EEXI requires the documents on **speed-power curve and tank test result** provided by mother shipyard and **SFC (recorded at NOx measurement)** provided by engine manufacturer.

Engine Power Limitation (EPL)

- **EEXI assessment beforehand is important** to find the impact on ship's operation as **ship's maximum speed will be reduced** due to EPL.
- **Installation of log recording device** may be required due to EPL.
- **Preparing beforehand** is recommended to avoid **congestion of EPL works** as EEXI regulation applies to the existing ships all over the world.

EEXI (Energy Efficiency eXisting ship Index)

Points of attention for EEXI (2/2)

ClassNK

EEXI verification by Class

- **Class approval** of EEXI technical file and EPL onboard management manual is required.
- **Some time is needed for review** to confirm the evidence of speed-power curve if ship's speed is calculated by speed-power curve.
- Drawing approval and onboard inspection by class **before 1st January 2023 is possible** although EEXI regulation will take place at the first annual, intermediate or renewal survey of IAPP Certificate on or after 1st January 2023.

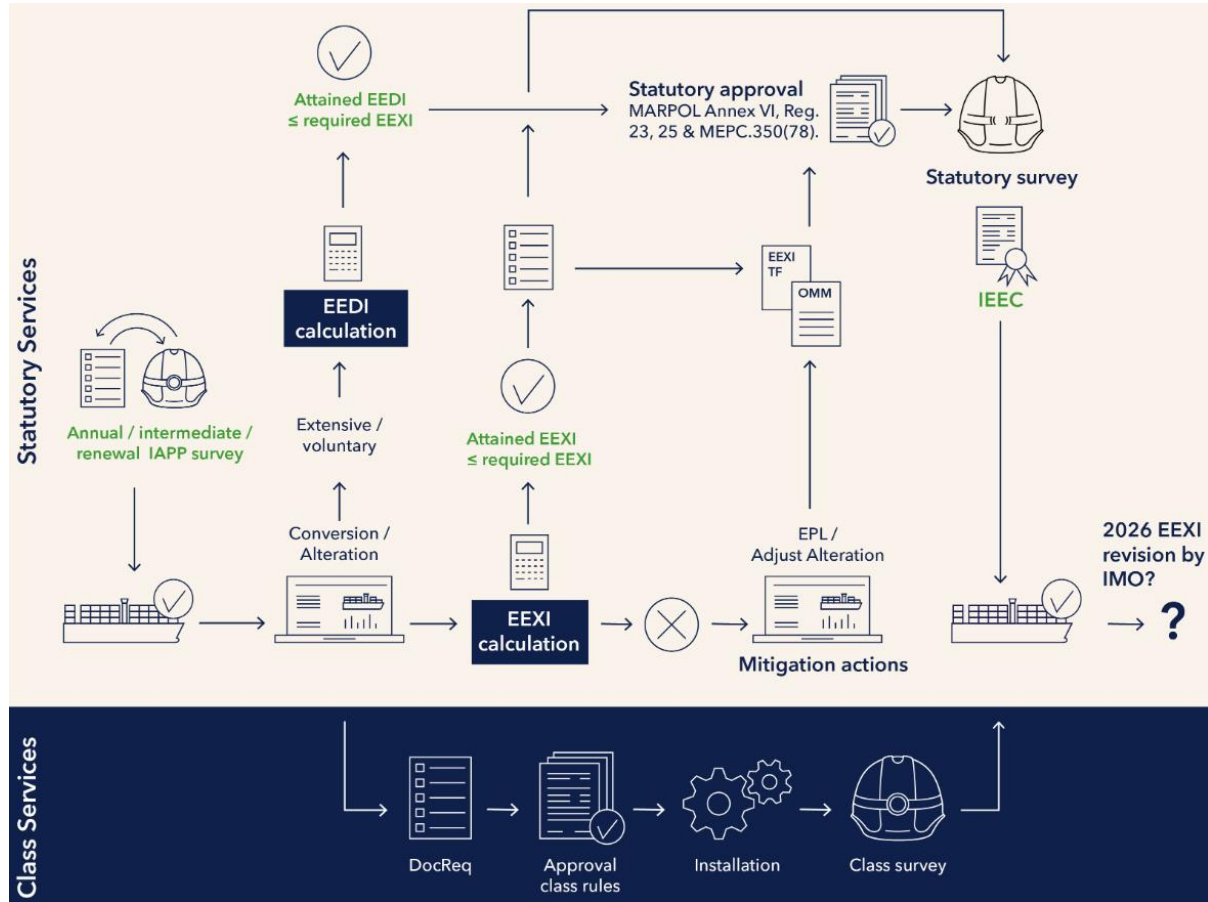
Cost for conformation to EEXI regulation

- Fee for making EEXI technical file and EPL onboard management manual
- Fee for data of speed-power curve, tank test result, SFC
- Fee for EPL setting
- Fee for EEXI verification by class (drawing approval, onboard inspection, issuance of new IEE certificate)



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EEXI (Energy Efficiency eXisting ship Index)



EEXI process for the fleet in service
(Figure is from DNV)

CII (Carbon Intensity Indicator)

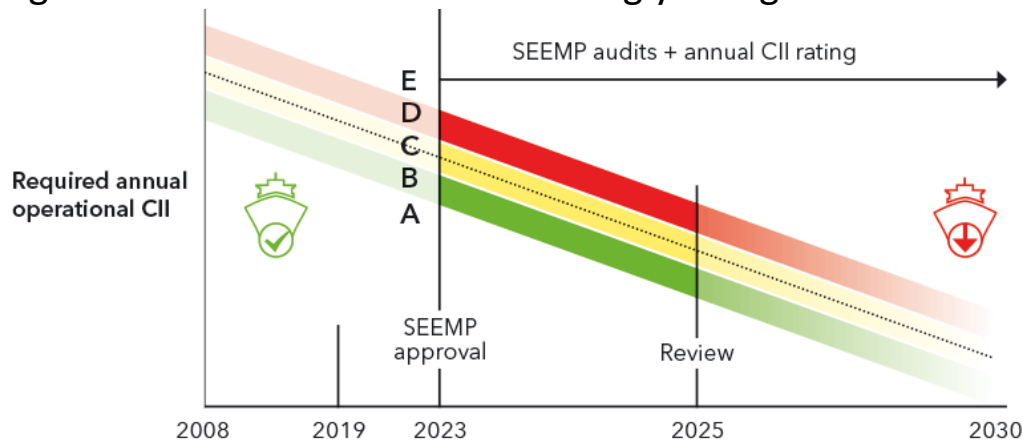
As a stimulus to reduce carbon intensity of all ships by at least 40% by 2030 compared to 2008 baseline, ships are required to calculate two ratings: their attained Energy Efficiency Existing Ship Index (EEXI) to determine their energy efficiency, and their annual operational Carbon Intensity Indicator (CII) and associated CII rating. Carbon intensity links the GHG emissions to the amount of cargo carried over distance travelled.

CII (Carbon Intensity Indicator)

The CII is a measure for a ship's energy efficiency and is given in grams of CO₂ emitted per cargo-carrying capacity and nautical mile.

This is a mandatory measure under MARPOL Annex VI, which came into force on the 1st January 2023. (CII is calculated based on a vessel's mandatory annual fuel report to the IMO.)

The first year of the attained annual operational CII verification will be 2024 for the operation in calendar year 2023. Vessels, based on their performance, will receive an environmental rating of A (major superior), B (minor superior), C (moderate), D (minor inferior) or E (inferior performance level). The rating thresholds will become increasingly stringent towards 2030.



CII (Carbon Intensity Indicator)

The measure impacts all cargo, RoPax and cruise vessels above 5,000 gross tonnage (GT) and trading internationally.

Within three months after the end of each calendar year, ships are required to report to its flag Administration, or any RO (Recognized Organization = generally classification society) duly authorized by it, the attained annual operational CII.

The CII determines the annual reduction factor needed to continuously improve the ship's operational carbon intensity within a specific rating level. The actual annual operational CII achieved will need to be documented and verified against the required annual operational CII.

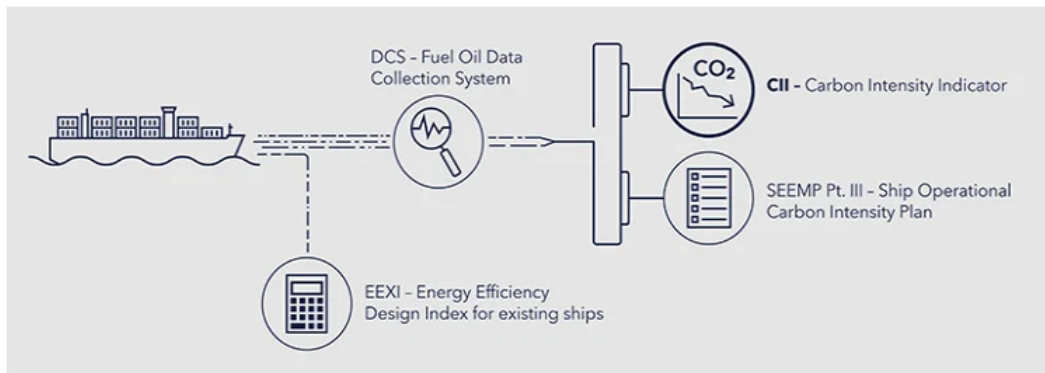
CII (Carbon Intensity Indicator)

This will then enable the operational carbon intensity rating to be determined on a scale of A, B, C, D or E, indicating a major superior, minor superior, moderate, minor inferior, or inferior performance level. The performance level will need to be recorded in the ship's Ship Energy Efficiency Management Plan (SEEMP). A ship rated D or E for three consecutive years will need to submit a corrective action plan to show how the required index rating (C or above) will be obtained.

This short-term measure aims to achieve the carbon intensity reduction aims of the IMO's initial GHG emissions strategy.

CII (Carbon Intensity Indicator)

Connection between the CII, DCS and SEEMP Part III



Starting in 2024, the CII must be calculated and reported to the Data Collection System (DCS) verifier together with the aggregated DCS data for the previous year, including any correction factors and voyage adjustments. Deadline for DCS and CII submission remains unchanged - no later than 31 March each year.

The attained annual operational CII and the environmental rating (A to E) will be noted on the DCS Statement of Compliance (SoC), which will be required to be kept on board for five years.

In case of a D rating for three consecutive years or one E rating, the SEEMP Part III must be updated with a corrective action plan and verified before the SoC can be issued. The corrective action plan should consist of an analysis of why the required CII was not achieved and include a revised implementation plan.



CII (Carbon Intensity Indicator)

Calculation of the CII

The CII unit is “grams of CO₂ emitted per cargo-carrying capacity and nautical mile”, whereby cargo capacity is either deadweight or gross tonnage depending on ship type. In addition, to cater for special design and operational circumstances, the correction factors and voyage adjustments can be applied to the basic CII calculations for the purposes of determining the rating.

Simplified attained annual CII formula:

$$\text{CII} = \frac{\text{Annual fuel consumption} \cdot \text{CO}_2 \text{ factor}}{\text{Annual distance travelled} \cdot \text{Capacity}} \cdot \text{Correction factors}$$



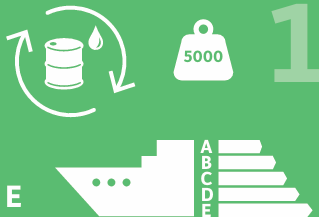
CII (Carbon Intensity Indicator)

CARBON INTENSITY INDICATOR (CII RATING)



IMPROVING THE OPERATIONAL PERFORMANCE OF EXISTING SHIPS

Each year, ships of 5,000 gross tonnage and above **collect and report fuel consumption data**. On the basis of this data, **A CARBON INTENSITY RATING IS ASSIGNED TO THE SHIP, FROM A TO E**



There are a variety of operational means to **IMPROVE THE CARBON INTENSITY OF EXISTING SHIPS** and achieve the Required CII, e.g.:

- Ship speed optimization
- Weather routing
- Just-in-time arrival
- Trim, draft, and ballast optimization



Poorly rated ships **have to implement A PLAN OF CORRECTIVE ACTIONS**, and the company is regularly audited incentives may be provided to best rated (A/B) ships



The requirements for CII rating ENTERED INTO EFFECT on 1 January 2023



Administrations, port authorities and other stakeholders as appropriate, are encouraged to provide incentives to ships rated as A or B.



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IMO DCS (Data Collection System)

In October 2016, MEPC 70 adopted, by resolution MEPC.278(70), the IMO ship fuel oil consumption system (IMO DCS), consisting of requirements for ships to record and report their fuel oil consumption with a view to inform further IMO measures to reduce GHG emissions from ships.

Starting from 1 January 2019, ships of 5,000 GT and above (which produce approximately 85% of the total CO₂ emissions from international shipping) are required to collect consumption data for each type of fuel oil they use as well as other specified data.

Since 2023, IMO DCS data is used to calculate ship's operational carbon intensity (CII).

IMO DCS

STANDARDIZED DATA REPORTING FORMAT
FOR THE DATA COLLECTION SYSTEM

Start date (dd/mm/yyyy)		01/01/2019
End date (dd/mm/yyyy)		31/12/2019
IMO number		IMO 9123348
Ship type		Chemical/Products Tanker
Gross tonnage		5,979
NT		4,123
DWT		10,115
EEDI (if applicable) (gCO ₂ /t.nm)		123.123456
Ice class (if applicable)		IA
Power output (rated power) (kW)	Main Propulsion Power	7,360
	No. 1 Auxiliary Engine	901
	No. 2 Auxiliary Engine	789
	No. 3 Auxiliary Engine	456
	No. 4 Auxiliary Engine	123
	(추가 계속)	
Distance Travelled (nm)		999,999,999,999
Hours underway (h)		7,300
Fuel oil consumption (t)	Diesel/Gas Oil (Cf : 3.206)	500
	LFO (Cf: 3.151)	1,000
	HFO (Cf: 3.114)	12,000
	LPG (Propane)	0
	LPG (Butane) (Cf: 3.030)	0
	LNG (Cf: 2.750)	0
	Methanol (Cf: 1.375)	0
	Ethanol (Cf: 1.913)	0
	Other(.....) (Cf :.....)	
Method used to measure uel oil consumption		1



Scrubber - SOx

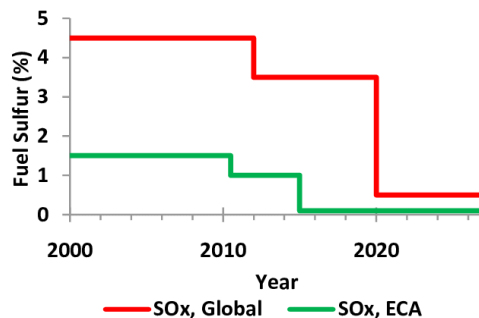
Exhaust gas cleaning systems (EGCS), or scrubbers, are becoming a more frequently used technology for ships to achieve compliance with the 2020 sulphur cap.

A scrubber sprays seawater or fresh water mixed with a caustic chemical into the exhaust gas stream in several stages.

The pollutant – mainly sulphur dioxide – reacts with the alkaline water, forming sulphuric acid.

In the case of an open-loop system, the resulting wash water is discharged back into the sea.

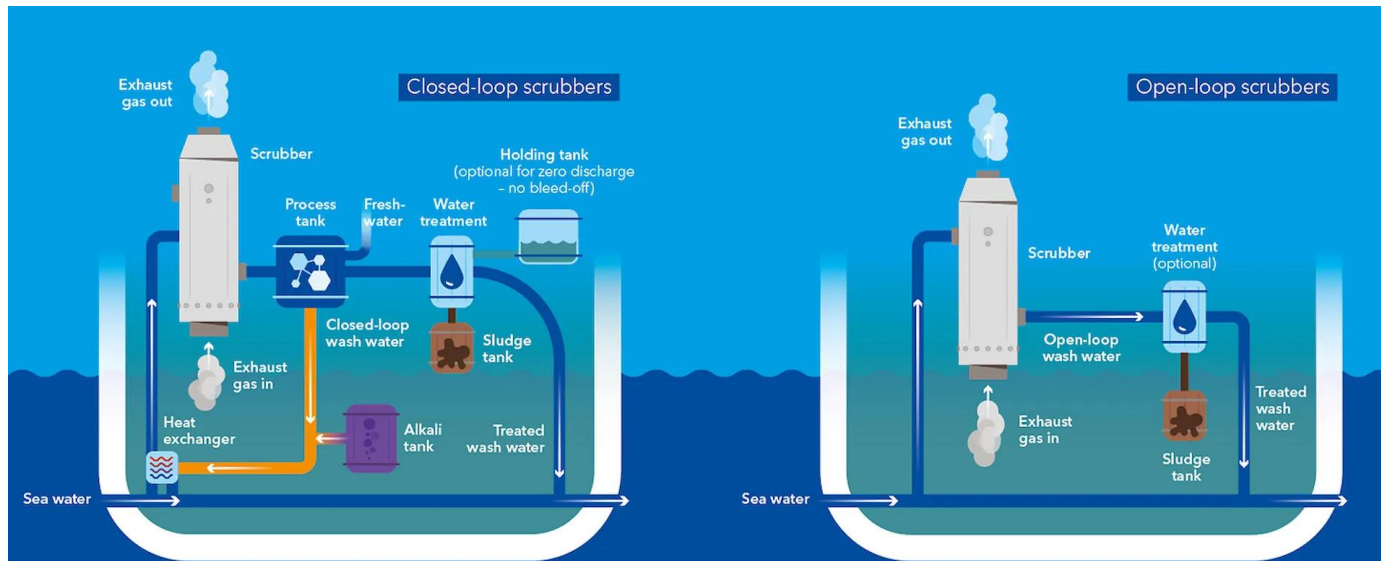
In areas and ports where open-loop scrubbing is prohibited, ships can use closed-loop systems and collect the accumulated sludge on board for subsequent disposal at a suitable in-port facility.



Scrubber

Hybrid scrubber systems can operate either in closed-loop or in open-loop mode, offering more flexibility.

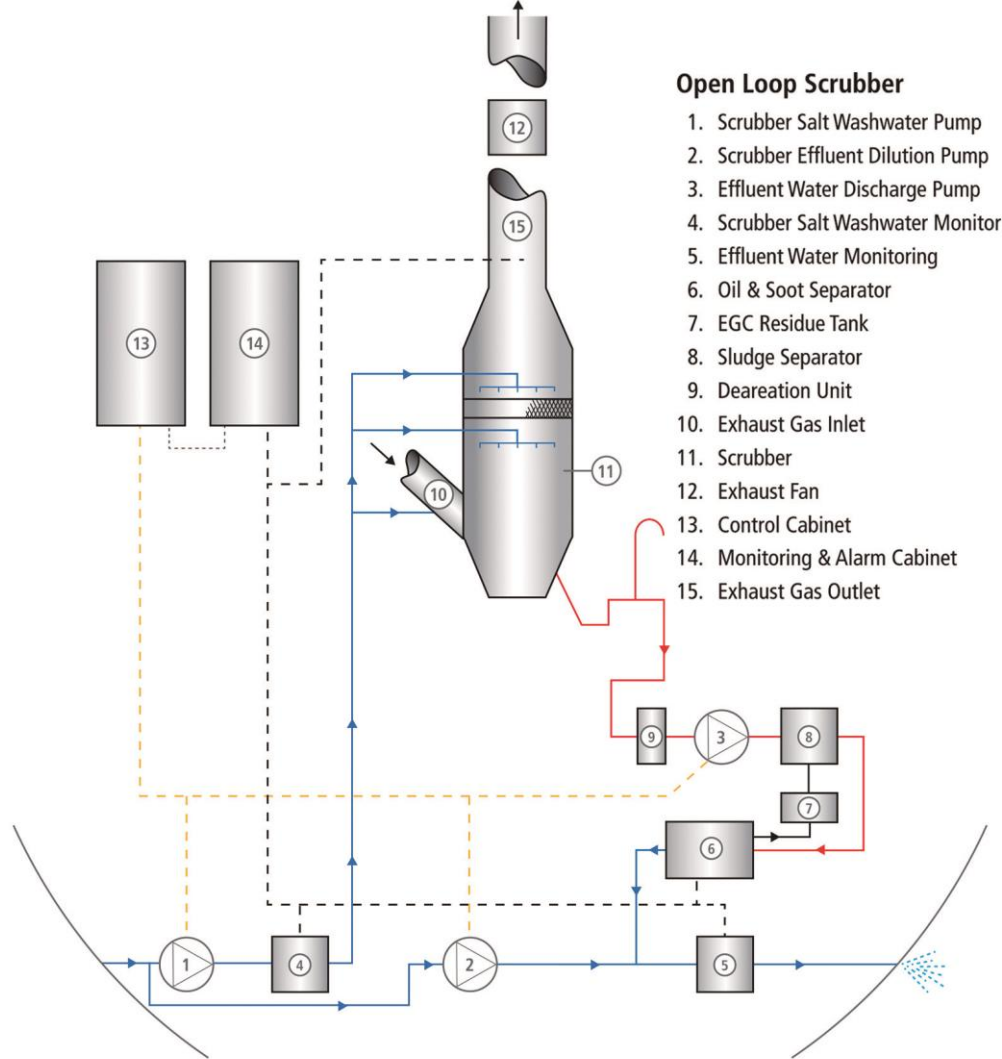
A multi-inlet scrubber can treat the exhaust gas from several engines, whereas a single-inlet scrubber serves one engine only.



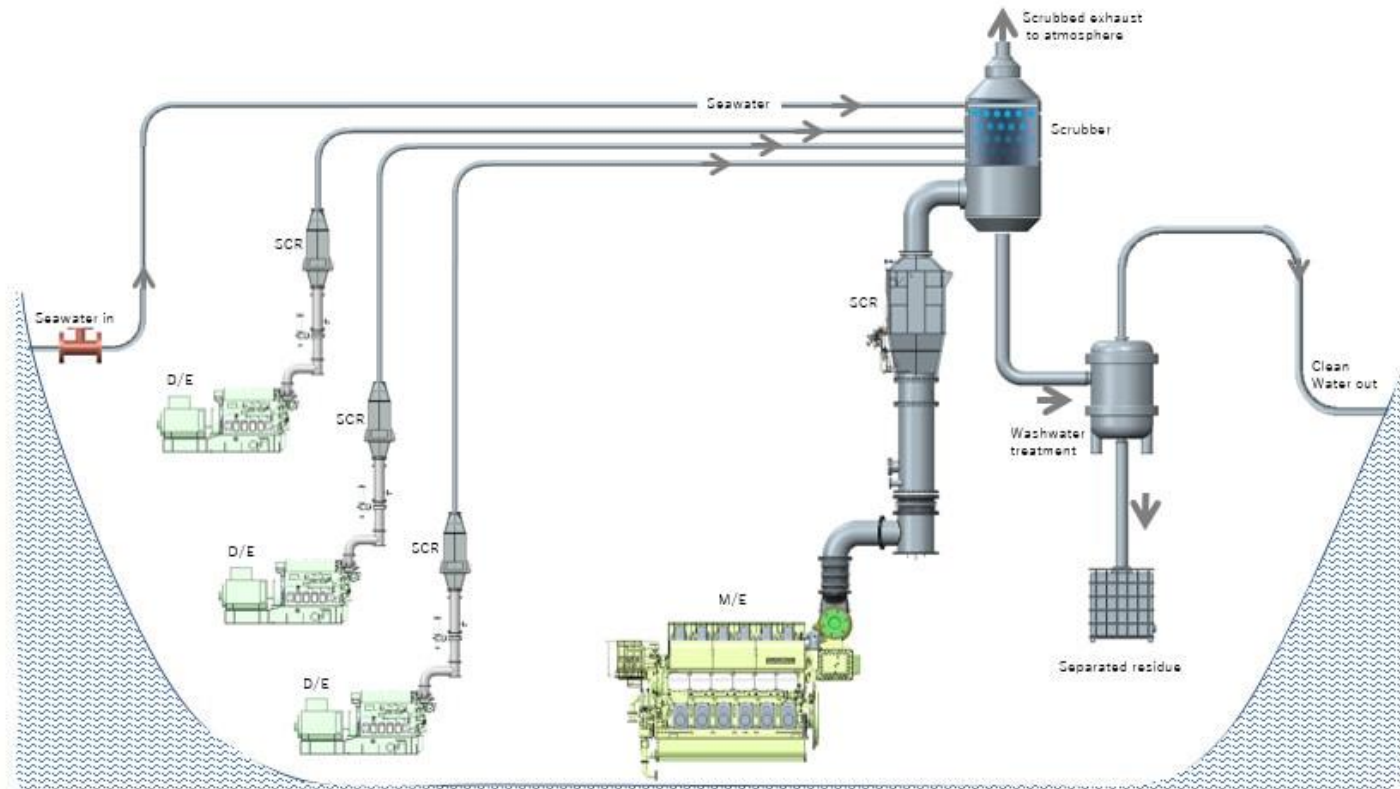
Scrubber

Closed-loop systems add chemicals, such as caustic soda, to the wash water to boost its alkalinity. The wash water is then recirculated through the system and partially purged.

Open-loop systems use seawater, which is alkaline by nature, to wash the SOx out of the exhaust. The resulting discharge water must meet MARPOL requirements before being discharged.



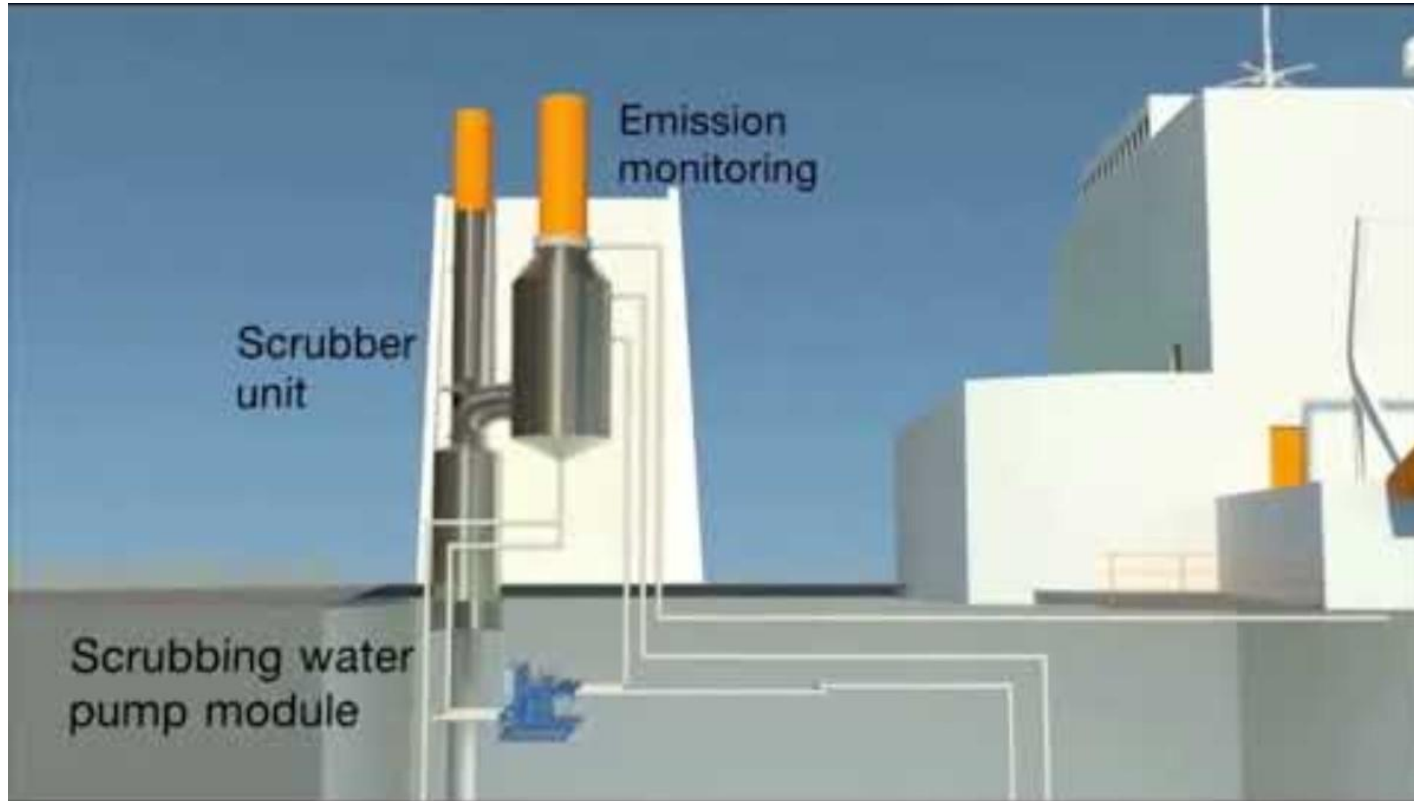
Scrubber



Scrubber



Scrubber



EGR (Exhaust gas recirculation) - NO_x

Nitrogen oxides (NO_x) emission reduction technique used in a reciprocating internal combustion engine, which involves diluting the intake air with recirculated exhaust gases.

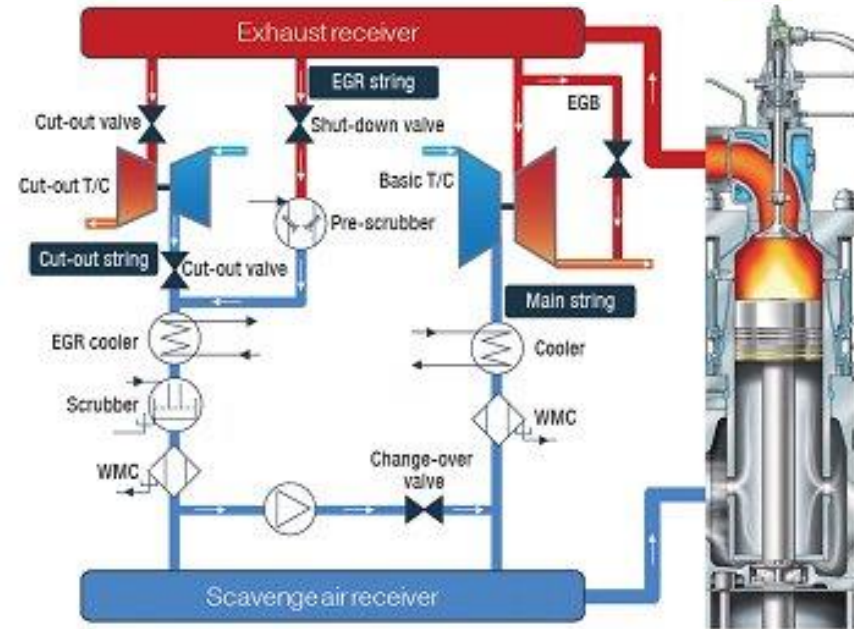
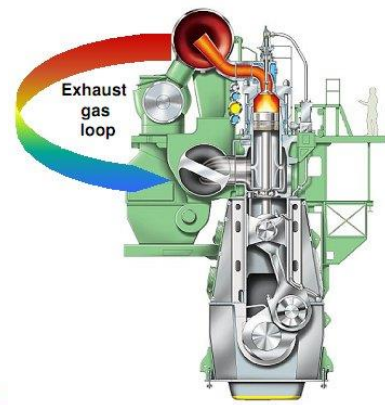
When measured in the exhaust duct of a marine diesel engine nitrogen oxide or NO_x emissions comprise about 95% nitric oxide (NO) and 5% nitrogen dioxide (NO₂).

The formation rate of the majority of nitric oxide (NO) is dependent on peak temperatures in the engine cylinders – above 1200°C the formation is significant and above 1500°C it becomes rapid.

EGR (Exhaust gas recirculation)

A proportion of the exhaust gas from the main engine exhaust receiver is recirculated to the scavenge air via a closed loop scrubber which removes damaging contaminants (PM, sulphur oxides etc.)

- O_2 in the scavenge air is replaced with CO_2 .
- CO_2 has a higher heat capacity than O_2 which reduces peak temperatures in the cylinder.
- Reduced O_2 content in the scavenge air reduces the combustion speed which reduces peak temperatures in the cylinder.
- Lower peak temperatures reduce the formation of NO and therefore NO_x for Tier III compliance.



EGR (Exhaust gas recirculation)

EGR inlet pipe &
pre-scrubber (not seen)

Distribution chamber

EGR cooler

EGR blower

EGR scrubber

Water Mist Catcher

EGR mixing

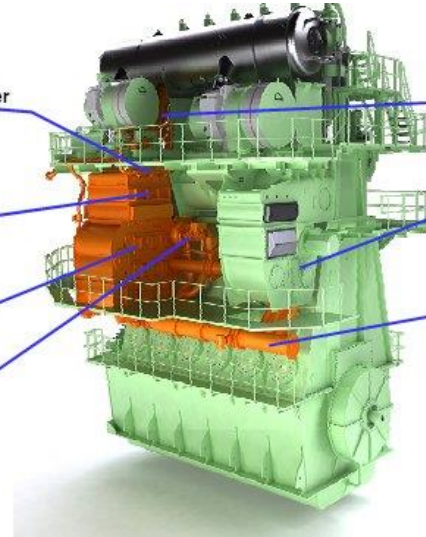


Distribution chamber

Dual cooler

EGR scrubber

EGR blower



EGR inlet pipe &
pre-scrubber

Mixing chamber

EGR outlet pipe

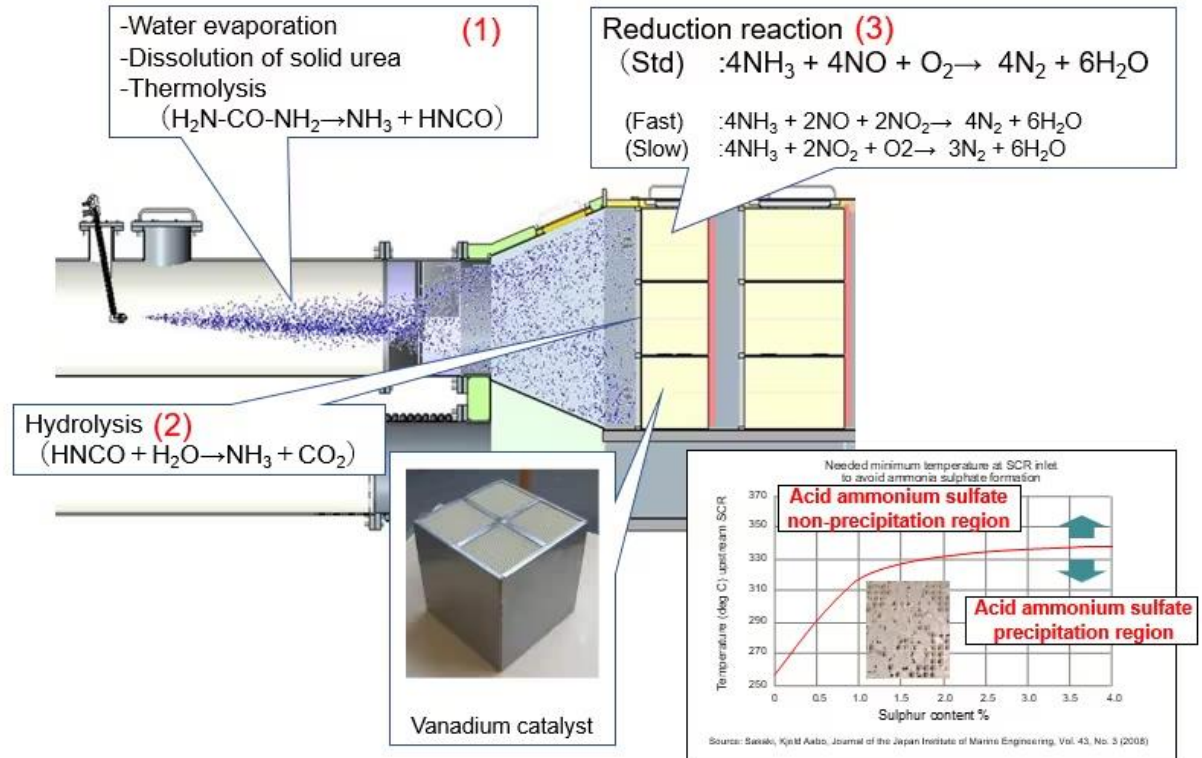
EGR integrated engine design MAN B&W 6S80ME-C9.2

SCR (Selective Catalytic Reduction) - NOx

SCR uses urea water, (solid urea ($\text{H}_2\text{N}-\text{CO}-\text{NH}_2$) dissolved in water) as its reducing agent.

When urea water is injected into exhaust gas, a process of water evaporation $\rightarrow$ dissolution of solid urea $\rightarrow$ thermolysis occurs.

Furthermore, hydrolysis of isocyanic acid (HNCO) occurs on the catalyst ((2)), resulting in the production of 2 mols of NH_3 from 1 mol of urea ((1) + (2)).

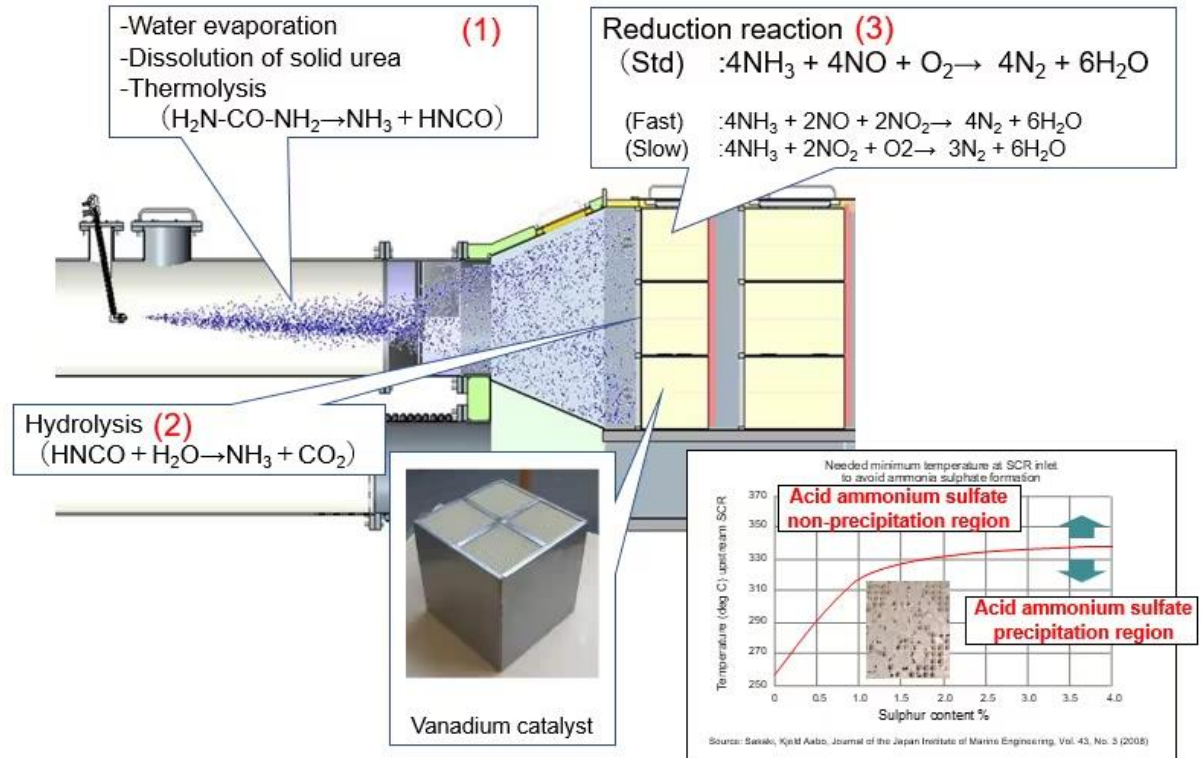


SCR (Selective Catalytic Reduction)

On the catalyst, the NH_3 selectively reduces the NO_x , breaking it down into harmless nitrogen and water.

The reduction reaction occurring on the catalyst can be standard, fast, or slow, depending on the NO/NO_2 ratio of the NO_x .

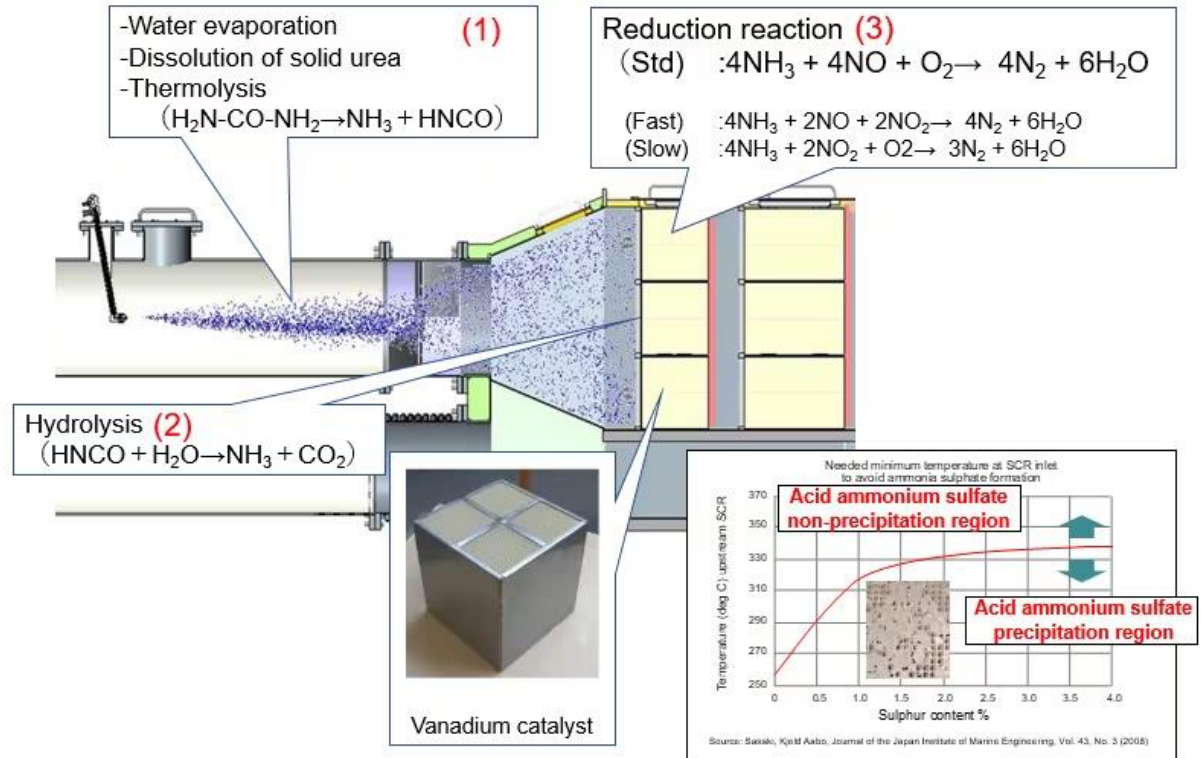
For medium- to high-speed marine engines, the majority of NO_x is NO , so the standard reaction is dominant in almost all cases ((3)).



SCR (Selective Catalytic Reduction)

There are multiple SCR catalysts available, each with a different main chemical components.

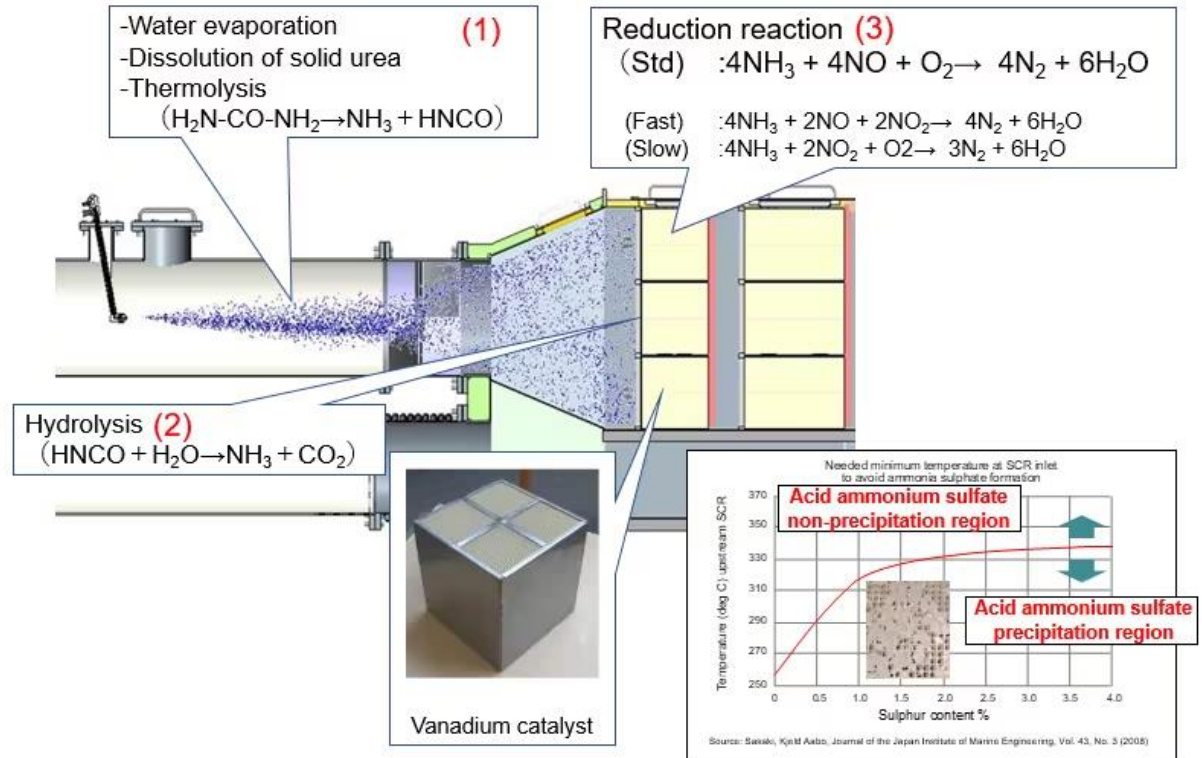
In the case of marine engines, which use fuel with a relatively high sulfur concentration, vanadium-based catalysts are used because of their resistance to sulfur poisoning.



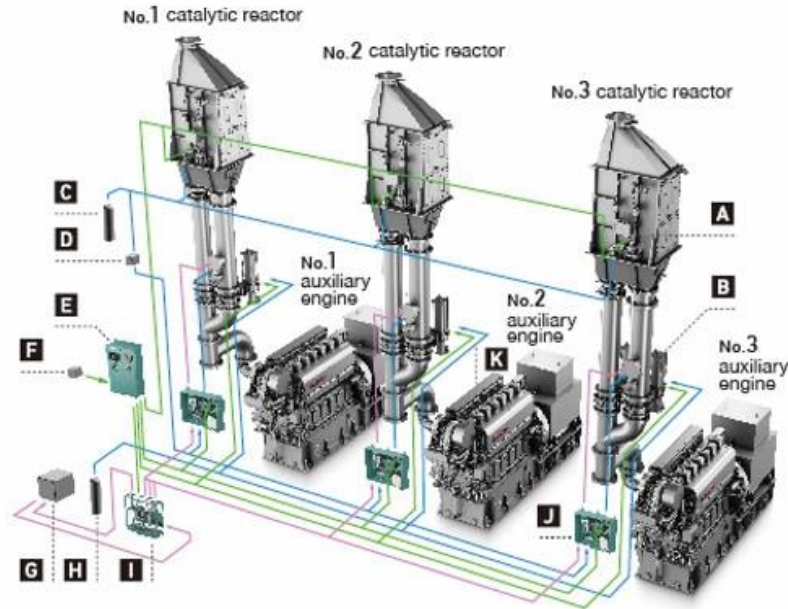
SCR (Selective Catalytic Reduction)

With SCR, there exists a region in the relationship between fuel sulfur concentration and exhaust gas temperature where acidic ammonium sulfate precipitation occurs, as shown in the graph in figure.

Because acidic ammonium sulfate clogs the catalyst and lowers the NO_x reduction rate, it is necessary to keep the exhaust gas at an appropriate temperature (at the SCR inlet).



SCR (Selective Catalytic Reduction)



— Air — Electricity — Urea water solution

A Soot blower

B Urea injection nozzle

C Service air

D Separator

E System control panel

F Temperature and humidity sensor

G Urea water tank

H Control air

I Pump unit

J Nozzle unit

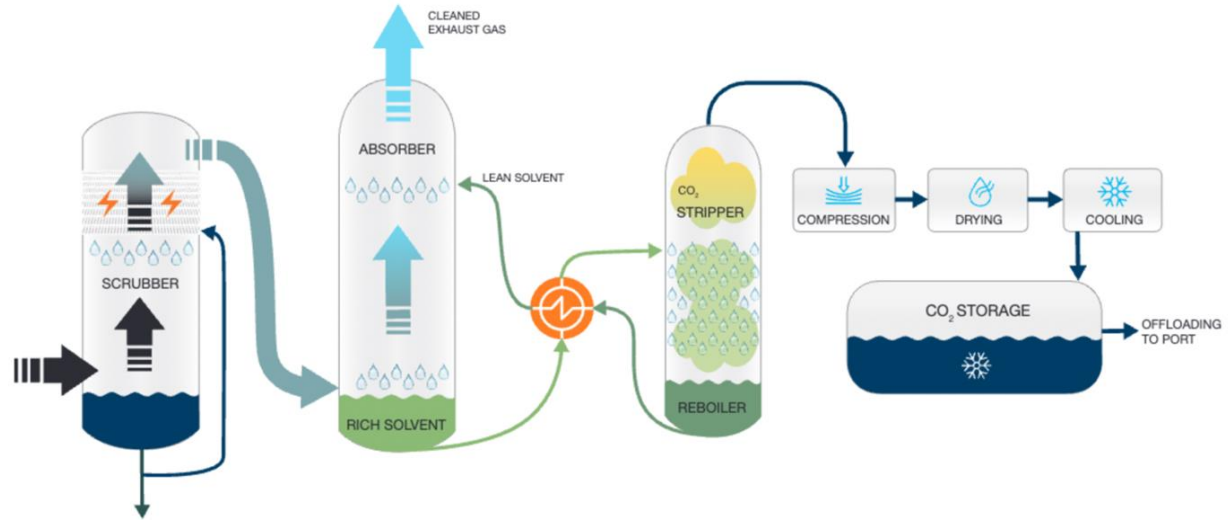
K Relief valve of boost air



CCS (Carbon capture and storage) - CO₂

Carbon capture is a method of capturing CO₂ from fossil fuel after it is combusted in an engine, before it is released into the atmosphere. Wärtsilä has developed an industry-first full-scale onboard carbon capture system which has the potential to reduce a vessel's CO₂ emissions by up to 70%.

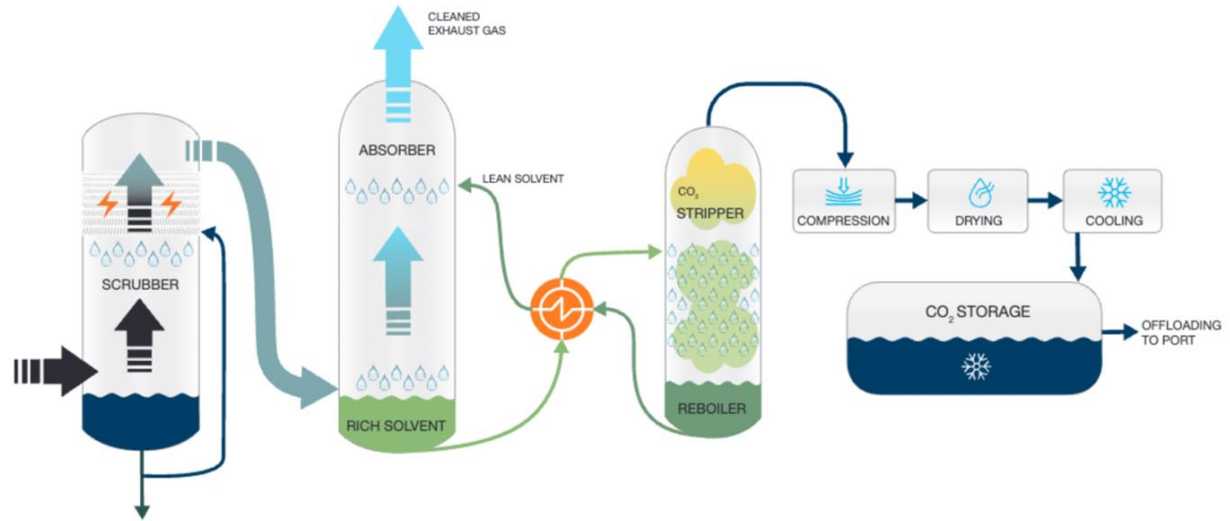
The Wärtsilä carbon capture solution works in five main stages: pre-conditioning, absorption, desorption (or stripping), liquefaction and storage.



CCS (Carbon capture and storage)

In pre-conditioning – depending on the fuel, the exhaust gas is cleaned of NO_x, SO_x and particulate matter and cooled down.

In the second stage, the exhaust gas is mixed with a liquid solvent. This captures the CO₂ in the solvent. The cleaned exhaust gas exits the system.

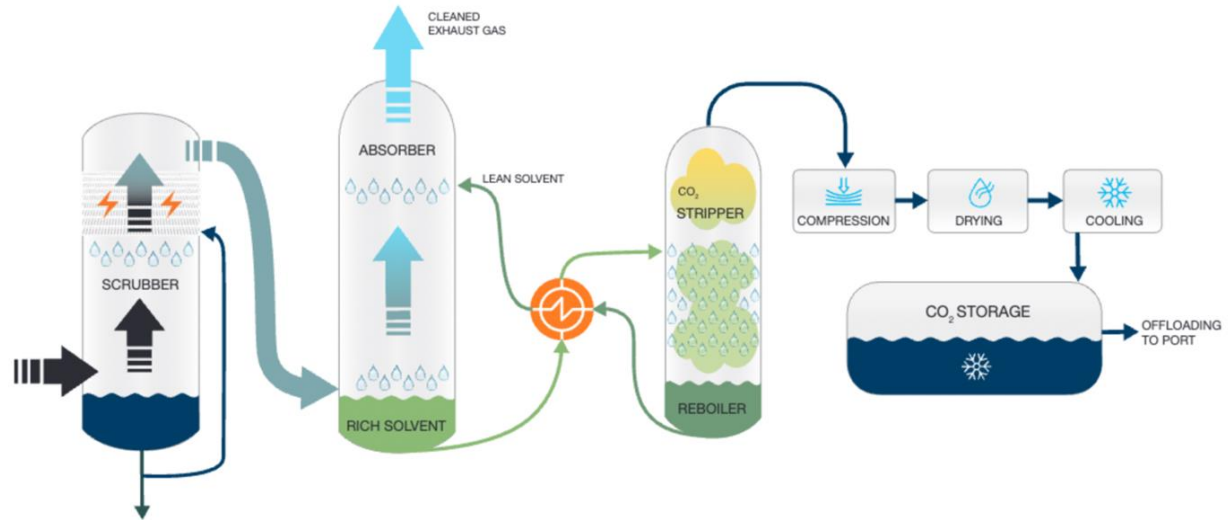


CCS (Carbon capture and storage)

In the stripping stage, the CO_2 -saturated solvent is heated which releases the CO_2 . The lean solvent will be reused in the system.

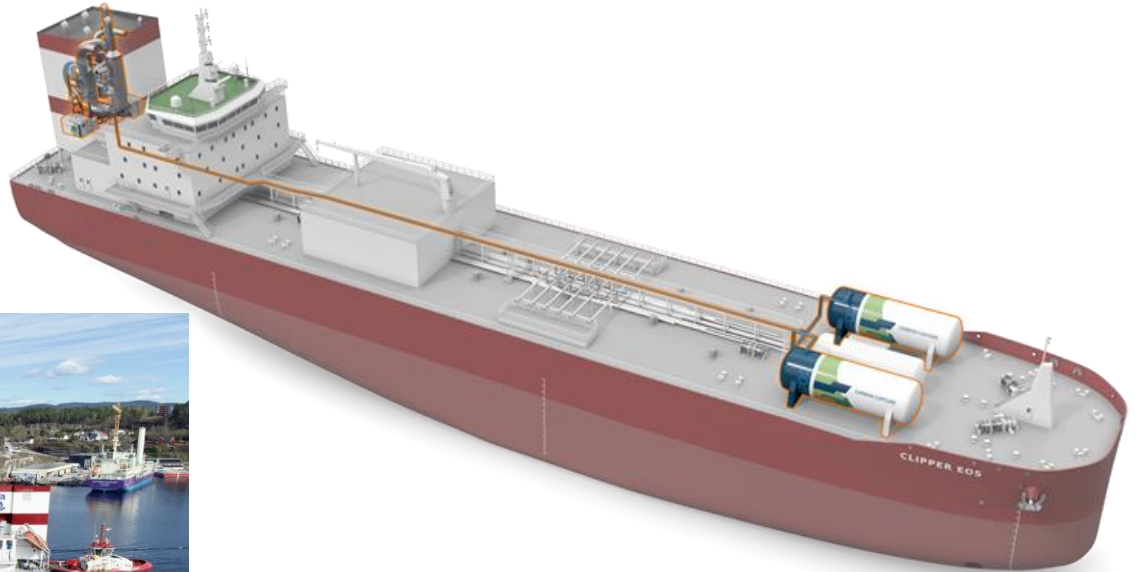
In the liquefaction stage, the CO_2 is compressed, dried and cooled down to liquid form for storage.

The final stage is storage. The liquid CO_2 is pumped through pipes to onboard tank(s), where it is stored until it can be offloaded.



CCS (Carbon capture and storage)

Clipper Eris, an ethylene carrier operated by the Norwegian Solvang Shipping is sailing with an onboard CCS system in the world-first full-scale maritime pilot.



CCS (Carbon capture and storage)



The OCCS-chimney is being hoisted on board Clipper Eris at Seatrume yard, Singapore.



CCS (Carbon capture and storage)

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CCS (Carbon capture and storage)



CCS (Carbon capture and storage)



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